

AWS Certified Solutions Architect - Associate SAA-C03

question 0101-0200

Last Update: 2024.11

【題號】question 101

【題目】

A solutions architect is designing a VPC with public and private subnets. The VPC and subnets use IPv4 CIDR blocks. There is one public subnet and one private subnet in each of three Availability Zones (AZs) for high availability. An internet gateway is used to provide internet access for the public subnets. The private subnets require access to the internet to allow Amazon EC2 instances to download software updates.

What should the solutions architect do to enable Internet access for the private subnets?

解決方案架構師正在設計具有公有子網路和私有子網路的 VPC。VPC 和子網路使用 IPv4 CIDR 區塊。三個可用區 (AZ) 中各有 1 個公有子網和 1 個私有子網，以實現高可用性。Internet 閘道用於為公用子網路提供 Internet 存取。私有子網路需要存取網際網路才能允許 Amazon EC2 執行個體下載軟體更新。

解決方案架構師應該怎麼做才能實現私有子網路的 Internet 存取？

【選項】

A.Create three NAT gateways, one for each public subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT gateway in its AZ.

B.Create three NAT instances, one for each private subnet in each AZ. Create a private route table for each AZ that forwards non-VPC traffic to the NAT instance in its AZ.

C.Create a second internet gateway on one of the private subnets. Update the route table for the private subnets that forward non-VPC traffic to the private internet gateway.

D.Create an egress-only internet gateway on one of the public subnets. Update the route table for the private subnets that forward non-VPC traffic to the egress-only Internet gateway.

【參考答案】A

【題號】question 102

【題目】

A company wants to migrate an on-premises data center to AWS. The data center hosts an SFTP server that stores its data on an NFS-based file system. The server holds 200 GB of data that needs to be transferred. The server must be hosted on an Amazon EC2 instance that uses an Amazon Elastic File System (Amazon EFS) file system.

Which combination of steps should a solutions architect take to automate this task? (Choose two.)

一家公司希望將本地資料中心遷移到 AWS。資料中心託管一個 SFTP 伺服器，該伺服器將其資料儲存在基於 NFS 的檔案系統上。伺服器保存需要傳輸的 200 GB 資料。伺服器必須託管在使用 Amazon Elastic File System (Amazon EFS) 檔案系統的 Amazon EC2 執行個體上。

解決方案架構師應採取哪些步驟組合來自動執行此任務？（選兩個。）

【選項】

- A. Launch the EC2 instance into the same Availability Zone as the EFS file system.
- B. Install an AWS DataSync agent in the on-premises data center.
- C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.
- D. Manually use an operating system copy command to push the data to the EC2 instance.
- E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server.

【參考答案】BE

【題號】question 103

【題目】

A company has an AWS Glue extract, transform, and load (ETL) job that runs every day at the same time. The job processes XML data that is in an Amazon S3 bucket. New data is added to the S3 bucket every day. A solutions architect notices that AWS Glue is processing all the data during each run.

What should the solutions architect do to prevent AWS Glue from reprocessing old data?

一家公司有一個每天同時運行的 AWS Glue 提取、轉換和載入 (ETL) 作業。此作業處理 Amazon S3 儲存桶中的 XML 資料。每天都有新資料添加到 S3 儲存桶中。解決方案架構師注意到 AWS Glue 在每次運行期間處理所有資料。

解決方案架構師應該如何防止 AWS Glue 重新處理舊資料？

【選項】

- A.Edit the job to use job bookmarks.
- B.Edit the job to delete data after the data is processed.
- C.Edit the job by setting the NumberOfWorkers field to 1.
- D.Use a FindMatches machine learning (ML) transform.

【參考答案】A

【題號】question 104

【題目】

A solutions architect must design a highly available infrastructure for a website. The website is powered by Windows web servers that run on Amazon EC2 instances. The solutions architect must implement a solution that can mitigate a large-scale DDoS attack that originates from thousands of IP addresses. Downtime is not acceptable for the website.

Which actions should the solutions architect take to protect the website from such an attack? (Choose two.)

解決方案架構師必須為網站設計高度可用的基礎架構。本網站由在 Amazon EC2 執行個體上執行的 Windows Web 伺服器提供支援。解決方案架構師必須實施能夠緩解源自數千個 IP 地址的大規模 DDoS 攻擊的解決方案。網站無法接受停機時間。

解決方案架構師應該採取哪些措施來保護網站免受此類攻擊？（選兩個。）

【選項】

- A. Use AWS Shield Advanced to stop the DDoS attack.
- B. Configure Amazon GuardDuty to automatically block the attackers.
- C. Configure the website to use Amazon CloudFront for both static and dynamic content.
- D. Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs.
- E. Use EC2 Spot Instances in an Auto Scaling group with a target tracking scaling policy that is set to 80% CPU utilization.

【參考答案】AC

【題號】question 105

【題目】

A company is preparing to deploy a new serverless workload. A solutions architect must use the principle of least privilege to configure permissions that will be used to run an AWS Lambda function. An Amazon EventBridge (Amazon CloudWatch Events) rule will invoke the function.

Which solution meets these requirements?

一家公司正準備部署新的無伺服器工作負載。解決方案架構師必須使用最小權限原則來配置將用於執行 AWS Lambda 函數的權限。Amazon EventBridge (Amazon CloudWatch Events) 規則將呼叫該函數。

哪種解決方案滿足這些要求？

【選項】

A.Add an execution role to the function with `lambda:InvokeFunction` as the action and `*` as the principal.

B.Add an execution role to the function with `lambda:InvokeFunction` as the action and `Service: lambda.amazonaws.com` as the principal.

C.Add a resource-based policy to the function with `lambda:*` as the action and `Service: events.amazonaws.com` as the principal.

D.Add a resource-based policy to the function with `lambda:InvokeFunction` as the action and `Service: events.amazonaws.com` as the principal.

【參考答案】D

【題號】question 106

【題目】

A company is preparing to store confidential data in Amazon S3. For compliance reasons, the data must be encrypted at rest. Encryption key usage must be logged for auditing purposes. Keys must be rotated every year.

Which solution meets these requirements and is the MOST operationally efficient?

一家公司正準備將機密資料儲存在 Amazon S3 中。出於合規性原因，必須對靜態資料進行加密。出於審計目的，必須記錄加密金鑰的使用情況。鑰匙必須每年輪換一次。

哪種解決方案滿足這些要求且營運效率最高？

【選項】

A.Server-side encryption with customer-provided keys (SSE-C)

B.Server-side encryption with Amazon S3 managed keys (SSE-S3)

C.Server-side encryption with AWS KMS keys (SSE-KMS) with manual rotation

D.Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation

【參考答案】D

【題號】question 107

【題目】

A bicycle sharing company is developing a multi-tier architecture to track the location of its bicycles during peak operating hours. The company wants to use these data points in its existing analytics platform. A solutions architect must determine the most viable multi-tier option to support this architecture. The data points must be accessible from the REST API.

Which action meets these requirements for storing and retrieving location data?

一家自行車共享公司正在開發一種多層架構，以在高峰營運時間追蹤其自行車的位置。該公司希望在其現有的分析平台中使用這些數據點。解決方案架構師必須確定最可行的多層選項來支援該架構。資料點必須可從 REST API 存取。

哪種操作滿足儲存和檢索位置資料的這些要求？

【選項】

- A. Use Amazon Athena with Amazon S3.
- B. Use Amazon API Gateway with AWS Lambda.
- C. Use Amazon QuickSight with Amazon Redshift.
- D. Use Amazon API Gateway with Amazon Kinesis Data Analytics.

【參考答案】D

【題號】question 108

【題目】

A company has an automobile sales website that stores its listings in a database on Amazon RDS. When an automobile is sold, the listing needs to be removed from the website and the data must be sent to multiple target systems.

Which design should a solutions architect recommend?

一家公司擁有一個汽車銷售網站，該網站將其清單儲存在 Amazon RDS 的資料庫中。當汽車出售時，需要從網站上刪除列表，並且資料必須發送到多個目標系統。

解決方案架構師應該推薦哪種設計？

【選項】

A.Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) queue for the targets to consume.

B.Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) FIFO queue for the targets to consume.

C.Subscribe to an RDS event notification and send an Amazon Simple Queue Service (Amazon SQS) queue fanned out to multiple Amazon Simple Notification Service (Amazon SNS) topics. Use AWS Lambda functions to update the targets.

D.Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

【參考答案】A

【題號】question 109

【題目】

A company needs to store data in Amazon S3 and must prevent the data from being changed. The company wants new objects that are uploaded to Amazon S3 to remain unchangeable for a nonspecific amount of time until the company decides to modify the objects. Only specific users in the company's AWS account can have the ability to delete the objects.

What should a solutions architect do to meet these requirements?

公司需要將資料儲存在 Amazon S3 中，並且必須防止資料被更改。該公司希望上傳到 Amazon S3 的新物件在一段非特定的時間內保持不變，直到公司決定修改這些物件。只有公司 AWS 帳戶中的特定使用者才可以刪除物件。10

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Create an S3 Glacier vault. Apply a write-once, read-many (WORM) vault lock policy to the objects.
- B.Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Set a retention period of 100 years. Use governance mode as the S3 bucket's default retention mode for new objects.
- C.Create an S3 bucket. Use AWS CloudTrail to track any S3 API events that modify the objects. Upon notification, restore the modified objects from any backup versions that the company has.
- D.Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Add a legal hold to the objects. Add the s3:PutObjectLegalHold permission to the IAM policies of users who need to delete the objects.

【參考答案】D

【題號】question 110

【題目】

A social media company allows users to upload images to its website. The website runs on Amazon EC2 instances. During upload requests, the website resizes the images to a standard size and stores the resized images in Amazon S3. Users are experiencing slow upload requests to the website.

The company needs to reduce coupling within the application and improve website performance. A solutions architect must design the most operationally efficient process for image uploads.

Which combination of actions should the solutions architect take to meet these requirements? (Choose two.)

一家社交媒體公司允許用戶將圖像上傳到其網站。網站在 Amazon EC2 執行個體上運作。在上傳請求期間，網站將圖像大小調整為標準大小並將調整大小的圖像儲存在 Amazon S3 中。使用者對網站的上傳請求緩慢。

該公司需要減少應用程式內的耦合並提高網站效能。解決方案架構師必須設計最高效的圖像上傳流程。

解決方案架構師應該採取哪些操作組合來滿足這些要求？（選兩個。）

【選項】

- A. Configure the application to upload images to S3 Glacier.
- B. Configure the web server to upload the original images to Amazon S3.
- C. Configure the application to upload images directly from each user's browser to Amazon S3 through the use of a presigned URL
- D. Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded. Use the function to resize the image.
- E. Create an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function on a schedule to resize uploaded images.

【參考答案】BD

【題號】question 111

【題目】

A company recently migrated a message processing system to AWS. The system receives messages into an ActiveMQ queue running on an Amazon EC2 instance. Messages are processed by a consumer application running on Amazon EC2. The consumer application processes the messages and writes results to a MySQL database running on Amazon EC2. The company wants this application to be highly available with low operational complexity.

Which architecture offers the HIGHEST availability?

一家公司最近將訊息處理系統遷移到 AWS。系統將訊息接收到在 Amazon EC2 執行個體上執行的 ActiveMQ 佇列中。訊息由在 Amazon EC2 上執行的消費者應用程式處理。消費者應

用程式處理訊息並將結果寫入 Amazon EC2 上執行的 MySQL 資料庫。該公司希望該應用程式具有高可用性和低操作複雜性。

哪種架構提供最高的可用性？

【選項】

A. Add a second ActiveMQ server to another Availability Zone. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

B. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

C. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Use Amazon RDS for MySQL with Multi-AZ enabled.

D. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an Auto Scaling group for the consumer EC2 instances across two Availability Zones. Use Amazon RDS for MySQL with Multi-AZ enabled.

【參考答案】D

【題號】question 112

【題目】

A company hosts a containerized web application on a fleet of on-premises servers that process incoming requests. The number of requests is growing quickly. The on-premises servers cannot handle the increased number of requests. The company wants to move the application to AWS with minimum code changes and minimum development effort.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司在一組處理傳入請求的本機伺服器上託管容器化 Web 應用程式。請求數量正在快速增加。本機伺服器無法處理增加的請求數量。該公司希望以最少的程式碼更改和最少的開發工作將應用程式遷移到 AWS。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A. Use AWS Fargate on Amazon Elastic Container Service (Amazon ECS) to run the containerized web application with Service Auto Scaling. Use an Application Load Balancer to distribute the incoming requests.

B. Use two Amazon EC2 instances to host the containerized web application. Use an Application Load Balancer to distribute the incoming requests.

C. Use AWS Lambda with a new code that uses one of the supported languages. Create multiple Lambda functions to support the load. Use Amazon API Gateway as an entry point to the Lambda functions.

D. Use a high performance computing (HPC) solution such as AWS ParallelCluster to establish an HPC cluster that can process the incoming requests at the appropriate scale.

【參考答案】A

【題號】question 113

【題目】

A company uses 50 TB of data for reporting. The company wants to move this data from on premises to AWS. A custom application in the company's data center runs a weekly data transformation job. The company plans to pause the application until the data transfer is complete and needs to begin the transfer process as soon as possible.

The data center does not have any available network bandwidth for additional workloads. A solutions architect must transfer the data and must configure the transformation job to continue to run in the AWS Cloud.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司使用 50 TB 的數據進行報告。該公司希望將這些資料從本地轉移到 AWS。公司資料中心的自訂應用程式每週執行一次資料轉換作業。該公司計劃暫停應用程式，直到數據傳輸完成，並需要盡快開始傳輸過程。

資料中心沒有任何可用的網路頻寬來處理額外的工作負載。解決方案架構師必須傳輸資料並配置轉換作業以繼續在 AWS 雲端中執行。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A. Use AWS DataSync to move the data. Create a custom transformation job by using AWS Glue.

B. Order an AWS Snowcone device to move the data. Deploy the transformation application to the device.

C. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue.

D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application.

【參考答案】C

【題號】question 114

【題目】

A company has created an image analysis application in which users can upload photos and add photo frames to their images. The users upload images and metadata to indicate which photo frames they want to add to their images. The application uses a single Amazon EC2 instance and Amazon DynamoDB to store the metadata.

The application is becoming more popular, and the number of users is increasing. The company expects the number of concurrent users to vary significantly depending on the time of day and day of week. The company must ensure that the application can scale to meet the needs of the growing user base.

Which solution meets these requirements?

一家公司創建了一個圖像分析應用程式，用戶可以在其中上傳照片並向圖像添加相框。用戶上傳圖像和元數據以指示他們想要將哪些相框添加到圖像中。該應用程式使用單一 Amazon EC2 執行個體和 Amazon DynamoDB 來儲存元資料。

該應用程式越來越受歡迎，用戶數量也在增加。該公司預計並髮用戶數量會根據一天中的不同時間和一周中的不同日期而有很大差異。公司必須確保應用程式可以擴展以滿足不斷增長的用戶群的需求。

哪種解決方案滿足這些要求？

【選項】

- A. Use AWS Lambda to process the photos. Store the photos and metadata in DynamoDB.
- B. Use Amazon Kinesis Data Firehose to process the photos and to store the photos and metadata.
- C. Use AWS Lambda to process the photos. Store the photos in Amazon S3. Retain DynamoDB to store the metadata.
- D. Increase the number of EC2 instances to three. Use Provisioned IOPS SSD (io2) Amazon Elastic Block Store (Amazon EBS) volumes to store the photos and metadata.

【參考答案】C

【題號】question 115

【題目】

A medical records company is hosting an application on Amazon EC2 instances. The application processes customer data files that are stored on Amazon S3. The EC2 instances are hosted in public subnets. The EC2 instances access Amazon S3 over the internet, but they do not require any other network access.

A new requirement mandates that the network traffic for file transfers take a private route and not be sent over the internet.

Which change to the network architecture should a solutions architect recommend to meet this requirement?

一家醫療記錄公司正在 Amazon EC2 執行個體上託管一個應用程式。該應用程式處理儲存在 Amazon S3 上的客戶資料檔案。EC2 執行個體託管在公有子網路中。EC2 執行個體透過網際網路存取 Amazon S3, 但它們不需要任何其他網路存取。

新要求要求檔案傳輸的網路流量採用專用路由, 而非透過網路傳送。

解決方案架構師應該建議對網路架構進行哪些變更來滿足此要求?

【選項】

A.Create a NAT gateway. Configure the route table for the public subnets to send traffic to Amazon S3 through the NAT gateway.

B.Configure the security group for the EC2 instances to restrict outbound traffic so that only traffic to the S3 prefix list is permitted.

C.Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.

D.Remove the internet gateway from the VPC. Set up an AWS Direct Connect connection, and route traffic to Amazon S3 over the Direct Connect connection.

【參考答案】C

【題號】question 116

【題目】

A company uses a popular content management system (CMS) for its corporate website. However, the required patching and maintenance are burdensome. The company is redesigning its website and wants anew solution. The website will be updated four times a year and does not need to have any dynamic content available. The solution must provide high scalability and enhanced security.

Which combination of changes will meet these requirements with the LEAST operational overhead? (Choose two.)

一家公司為其公司網站使用流行的內容管理系統 (CMS)。然而, 所需的修補和維護是繁重的。該公司正在重新設計其網站並需要新的解決方案。網站每年更新四次, 不需要提供任何動態內容。該解決方案必須提供高可擴展性和增強的安全性。

哪種變更組合能夠以最少的營運開銷滿足這些要求？（選兩個。）

【選項】

- A. Configure Amazon CloudFront in front of the website to use HTTPS functionality.
- B. Deploy an AWS WAF web ACL in front of the website to provide HTTPS functionality.
- C. Create and deploy an AWS Lambda function to manage and serve the website content.
- D. Create the new website and an Amazon S3 bucket. Deploy the website on the S3 bucket with static website hosting enabled.
- E. Create the new website. Deploy the website by using an Auto Scaling group of Amazon EC2 instances behind an Application Load Balancer.

【參考答案】AD

【題號】question 117

【題目】

A company stores its application logs in an Amazon CloudWatch Logs log group. A new policy requires the company to store all application logs in Amazon OpenSearch Service (Amazon Elasticsearch Service) in near-real time.

Which solution will meet this requirement with the LEAST operational overhead?

本公司將其應用程式日誌儲存在 Amazon CloudWatch Logs 日誌組中。新政策要求該公司將所有應用程式日誌近乎即時地儲存在 Amazon OpenSearch Service (Amazon Elasticsearch Service) 中。

哪種解決方案能夠以最少的營運開銷滿足這項要求？

【選項】

- A. Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

B.Create an AWS Lambda function. Use the log group to invoke the function to write the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

C.Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery streams sources. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

D.Install and configure Amazon Kinesis Agent on each application server to deliver the logs to Amazon Kinesis Data Streams. Configure Kinesis Data Streams to deliver the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

【參考答案】A

【題號】question 118

【題目】

A company is building a web-based application running on Amazon EC2 instances in multiple Availability Zones. The web application will provide access to a repository of text documents totaling about 900 TB in size. The company anticipates that the web application will experience periods of high demand. A solutions architect must ensure that the storage component for the text documents can scale to meet the demand of the application at all times. The company is concerned about the overall cost of the solution.

Which storage solution meets these requirements MOST cost-effectively?

一家公司正在建立一個基於 Web 的應用程式，該應用程式在多個可用區的 Amazon EC2 執行個體上運行。此 Web 應用程式將提供對總計約 900 TB 大小的文字文件儲存庫的存取。該公司預計網路應用程式將經歷高需求時期。解決方案架構師必須確保文字文件的儲存組件可以擴展以滿足應用程式的需求。該公司擔心該解決方案的總體成本。

哪種儲存解決方案最經濟有效率地滿足這些要求？

【選項】

A.Amazon Elastic Block Store (Amazon EBS)

B.Amazon Elastic File System (Amazon EFS)

C.Amazon OpenSearch Service (Amazon Elasticsearch Service)

D.Amazon S3

【參考答案】D

【題號】question 119

【題目】

A global company is using Amazon API Gateway to design REST APIs for its loyalty club users in the us-east-1 Region and the ap-southeast-2 Region. A solutions architect must design a solution to protect these API Gateway managed REST APIs across multiple accounts from SQL injection and cross-site scripting attacks.

Which solution will meet these requirements with the LEAST amount of administrative effort?

一家跨國公司正在使用 Amazon API Gateway 為其位於 us-east-1 區域和 ap-southeast-2 區域的忠誠俱樂部使用者設計 REST API。解決方案架構師必須設計一個解決方案來保護這些跨多個帳戶的 API Gateway 管理的 REST API 免受 SQL 注入和跨站點腳本攻擊。

哪一種解決方案能夠以最少的管理工作來滿足這些要求？

【選項】

- A.Set up AWS WAF in both Regions. Associate Regional web ACLs with an API stage.
- B.Set up AWS Firewall Manager in both Regions. Centrally configure AWS WAF rules.
- C.Set up AWS Shield in both Regions. Associate Regional web ACLs with an API stage.
- D.Set up AWS Shield in one of the Regions. Associate Regional web ACLs with an API stage.

【參考答案】B

【題號】question 120

【題目】

A company has implemented a self-managed DNS solution on three Amazon EC2 instances behind a Network Load Balancer (NLB) in the us-west-2 Region. Most of the company's users are located in the United States and Europe. The company wants to improve the performance and availability of the solution. The company launches and configures three EC2 instances in the eu-west-1 Region and adds the EC2 instances as targets for a new NLB.

Which solution can the company use to route traffic to all the EC2 instances?

一家公司在 us-west-2 區域的網路負載平衡器 (NLB) 後面的三個 Amazon EC2 執行個體上實施了自我管理的 DNS 解決方案。該公司的大部分用戶位於美國和歐洲。該公司希望提高解決方案的效能和可用性。該公司在 eu-west-1 區域啟動並配置了三個 EC2 實例，並將這些 EC2 執行個體新增為新 NLB 的目標。

公司可以使用哪種解決方案將流量路由到所有 EC2 執行個體？

【選項】

A.Create an Amazon Route 53 geolocation routing policy to route requests to one of the two NLBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

B.Create a standard accelerator in AWS Global Accelerator. Create endpoint groups in us-west-2 and eu-west-1. Add the two NLBs as endpoints for the endpoint groups.

C.Attach Elastic IP addresses to the six EC2 instances. Create an Amazon Route 53 geolocation routing policy to route requests to one of the six EC2 instances. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

D.Replace the two NLBs with two Application Load Balancers (ALBs). Create an Amazon Route 53 latency routing policy to route requests to one of the two ALBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

【參考答案】B

【題號】question 121

【題目】

A company is running an online transaction processing (OLTP) workload on AWS. This workload uses an unencrypted Amazon RDS DB instance in a Multi-AZ deployment. Daily database snapshots are taken from this instance.

What should a solutions architect do to ensure the database and snapshots are always encrypted moving forward?

一家公司正在 AWS 上執行線上事務處理 (OLTP) 工作負載。此工作負載在多可用區部署中使用未加密的 Amazon RDS 資料庫執行個體。每日資料庫快照均從此實例中取得。

解決方案架構師應該做什麼來確保資料庫和快照始終加密？

【選項】

- A. Encrypt a copy of the latest DB snapshot. Replace existing DB instance by restoring the encrypted snapshot.
- B. Create a new encrypted Amazon Elastic Block Store (Amazon EBS) volume and copy the snapshots to it. Enable encryption on the DB instance.
- C. Copy the snapshots and enable encryption using AWS Key Management Service (AWS KMS) Restore encrypted snapshot to an existing DB instance.
- D. Copy the snapshots to an Amazon S3 bucket that is encrypted using server-side encryption with AWS Key Management Service (AWS KMS) managed keys (SSE-KMS).

【參考答案】A

【題號】question 122

【題目】

A company wants to build a scalable key management infrastructure to support developers who need to encrypt data in their applications.

What should a solutions architect do to reduce the operational burden?

一家公司希望建立可擴展的金鑰管理基礎設施來支援需要加密應用程式中的資料的開發人員。

解決方案架構師該怎麼做才能減輕營運負擔？

【選項】

- A. Use multi-factor authentication (MFA) to protect the encryption keys.
- B. Use AWS Key Management Service (AWS KMS) to protect the encryption keys.
- C. Use AWS Certificate Manager (ACM) to create, store, and assign the encryption keys.
- D. Use an IAM policy to limit the scope of users who have access permissions to protect the encryption keys.

【參考答案】B

【題號】question 123

【題目】

A company has a dynamic web application hosted on two Amazon EC2 instances. The company has its own SSL certificate, which is on each instance to perform SSL termination.

There has been an increase in traffic recently, and the operations team determined that SSL encryption and decryption is causing the compute capacity of the web servers to reach their maximum limit.

What should a solutions architect do to increase the application's performance?

一家公司有一個動態 Web 應用程式託管在兩個 Amazon EC2 執行個體上。該公司擁有自己的 SSL 證書，該證書在每個實例上執行 SSL 終止。

最近流量增加，營運團隊確定 SSL 加密和解密導致 Web 伺服器的運算能力達到最大限制。

解決方案架構師應該怎麼做才能提高應用程式的效能？

【選項】

A.Create a new SSL certificate using AWS Certificate Manager (ACM). Install the ACM certificate on each instance.

B.Create an Amazon S3 bucket Migrate the SSL certificate to the S3 bucket. Configure the EC2 instances to reference the bucket for SSL termination.

C.Create another EC2 instance as a proxy server. Migrate the SSL certificate to the new instance and configure it to direct connections to the existing EC2 instances.

D.Import the SSL certificate into AWS Certificate Manager (ACM). Create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM.

【參考答案】D

【題號】question 124

【題目】

A company has a highly dynamic batch processing job that uses many Amazon EC2 instances to complete it. The job is stateless in nature, can be started and stopped at any given time with no negative impact, and typically takes upwards of 60 minutes total to complete. The company has asked a solutions architect to design a scalable and cost-effective solution that meets the requirements of the job.

What should the solutions architect recommend?

一家公司有一個高度動態的批次作業，該作業使用許多 Amazon EC2 執行個體來完成。這份作業本質上是無狀態的，可以在任何給定時間啟動和停止，不會產生負面影響，並且通常總共需要 60 分鐘以上才能完成。該公司已要求解決方案架構師設計一個可擴展且經濟高效的解決方案，以滿足工作要求。

解決方案架構師該推薦什麼？

【選項】

A.Implement EC2 Spot Instances.

B.Purchase EC2 Reserved Instances.

C.Implement EC2 On-Demand Instances.

D.Implement the processing on AWS Lambda.

【參考答案】A

【題號】question 125

【題目】

A company runs its two-tier ecommerce website on AWS. The web tier consists of a load balancer that sends traffic to Amazon EC2 instances. The database tier uses an Amazon RDS DB instance. The EC2 instances and the RDS DB instance should not be exposed to the public internet. The EC2 instances require internet access to complete payment processing of orders through a third-party web service. The application must be highly available.

Which combination of configuration options will meet these requirements? (Choose two.)

一家公司在 AWS 上經營其兩層電子商務網站。Web 層由負載平衡器組成，將流量傳送至 Amazon EC2 執行個體。資料庫層使用 Amazon RDS 資料庫執行個體。EC2 執行個體和 RDS 資料庫執行個體不應暴露於公共網際網路。EC2 執行個體需要網際網路存取才能透過第三方 Web 服務完成訂單的付款處理。該應用程式必須具有高可用性。

哪種配置選項組合可以滿足這些要求？（選兩個。）

【選項】

A.Use an Auto Scaling group to launch the EC2 instances in private subnets. Deploy an RDS Multi-AZ DB instance in private subnets.

B.Configure a VPC with two private subnets and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the private subnets.

C.Use an Auto Scaling group to launch the EC2 instances in public subnets across two Availability Zones. Deploy an RDS Multi-AZ DB instance in private subnets.

D.Configure a VPC with one public subnet, one private subnet, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnet.D. Configure a VPC with two public subnets, two private subnets, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnets.

【參考答案】AD

【題號】question 126

【題目】

A solutions architect needs to implement a solution to reduce a company's storage costs. All the company's data is in the Amazon S3 Standard storage class. The company must keep all data for at least 25 years. Data from the most recent 2 years must be highly available and immediately retrievable.

Which solution will meet these requirements?

解決方案架構師需要實施一個解決方案來降低公司的儲存成本。本公司的所有資料都位於 Amazon S3 標準儲存類別中。公司必須將所有資料保存至少 25 年。最近 2 年的資料必須高度可用且可立即檢索。

哪種解決方案可以滿足這些要求？

【選項】

- A. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive immediately.
- B. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 2 years.
- C. Use S3 Intelligent-Tiering. Activate the archiving option to ensure that data is archived in S3 Glacier Deep Archive.
- D. Set up an S3 Lifecycle policy to transition objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately and to S3 Glacier Deep Archive after 2 years.

【參考答案】B

【題號】question 127

【題目】

A media company is evaluating the possibility of moving its systems to the AWS Cloud. The company needs at least 10 TB of storage with the maximum possible I/O performance for video processing, 300 TB of very durable storage for storing media content, and 900 TB of storage to meet requirements for archival media that is not in use anymore.

Which set of services should a solutions architect recommend to meet these requirements?

一家媒體公司正在評估將其系統遷移到 AWS 雲端的可能性。該公司需要至少10 TB 的儲存空間(具有最大可能的I/O 效能)用於視訊處理, 300 TB 的非常耐用的儲存空間用於儲存媒體內容, 以及900 TB 的儲存空間以滿足不再使用的存檔媒體的要求。

解決方案架構師應該推薦哪一組服務來滿足這些要求?

【選項】

- A.Amazon EBS for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage
- B.Amazon EBS for maximum performance, Amazon EFS for durable data storage, and Amazon S3 Glacier for archival storage
- C.Amazon EC2 instance store for maximum performance, Amazon EFS for durable data storage, and Amazon S3 for archival storage
- D.Amazon EC2 instance store for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage

【參考答案】D

【題號】question 128

【題目】

A company wants to run applications in containers in the AWS Cloud. These applications are stateless and can tolerate disruptions within the underlying infrastructure. The company needs a solution that minimizes cost and operational overhead.

What should a solutions architect do to meet these requirements?

一家公司希望在 AWS 雲端中的容器中運行應用程式。這些應用程式是無狀態的，可以容忍底層基礎設施內的中斷。該公司需要一種能夠最大限度地降低成本和營運開銷的解決方案。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A. Use Spot Instances in an Amazon EC2 Auto Scaling group to run the application containers.
- B. Use Spot Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group.
- C. Use On-Demand Instances in an Amazon EC2 Auto Scaling group to run the application containers.
- D. Use On-Demand Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group.

【參考答案】B

【題號】question 129

【題目】

A company is running a multi-tier web application on premises. The web application is containerized and runs on a number of Linux hosts connected to a PostgreSQL database that contains user records. The operational overhead of maintaining the infrastructure and capacity planning is limiting the company's growth. A solutions architect must improve the application's infrastructure.

Which combination of actions should the solutions architect take to accomplish this? (Choose two.)

一家公司正在本地運行多層 Web 應用程式。該 Web 應用程式是容器化的，並在連接到包含使用者記錄的 PostgreSQL 資料庫的多個 Linux 主機上運行。維護基礎設施和容量規劃的營運開銷限制了公司的成長。解決方案架構師必須改進應用程式的基礎架構。

解決方案架構師應該採取哪些行動組合來實現這個目標？（選兩個。）

【選項】

- A.Migrate the PostgreSQL database to Amazon Aurora.
- B.Migrate the web application to be hosted on Amazon EC2 instances.
- C.Set up an Amazon CloudFront distribution for the web application content.
- D.Set up Amazon ElastiCache between the web application and the PostgreSQL database.
- E.Migrate the web application to be hosted on AWS Fargate with Amazon Elastic Container Service (Amazon ECS).

【參考答案】AE

【題號】question 130

【題目】

An application runs on Amazon EC2 instances across multiple Availability Zones. The instances run in an Amazon EC2 Auto Scaling group behind an Application Load Balancer. The application performs best when the CPU utilization of the EC2 instances is at or near 40%.

What should a solutions architect do to maintain the desired performance across all instances in the group?

應用程式在跨多個可用區的 Amazon EC2 執行個體上執行。這些執行個體在 Application Load Balancer 後面的 Amazon EC2 Auto Scaling 群組中運作。當 EC2 執行個體的 CPU 使用率等於或接近 40% 時，應用程式效能最佳。

解決方案架構師應該怎麼做才能維持群組中所有實例的所需效能？

【選項】

- A. Use a simple scaling policy to dynamically scale the Auto Scaling group.
- B. Use a target tracking policy to dynamically scale the Auto Scaling group.
- C. Use an AWS Lambda function to update the desired Auto Scaling group capacity.
- D. Use scheduled scaling actions to scale up and scale down the Auto Scaling group.

【參考答案】B

【題號】question 131

【題目】

A company is developing a file-sharing application that will use an Amazon S3 bucket for storage. The company wants to serve all the files through an Amazon CloudFront distribution. The company does not want the files to be accessible through direct navigation to the S3 URL.

What should a solutions architect do to meet these requirements?

一家公司正在開發一個文件共享應用程式，該應用程式將使用 Amazon S3 儲存桶進行儲存。該公司希望透過 Amazon CloudFront 發行版提供所有文件。該公司不希望透過直接導航到 S3 URL 來存取這些文件。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A. Write individual policies for each S3 bucket to grant read permission for only CloudFront access.
- B. Create an IAM user. Grant the user read permission to objects in the S3 bucket. Assign the user to CloudFront.
- C. Write an S3 bucket policy that assigns the CloudFront distribution ID as the Principal and assigns the target S3 bucket as the Amazon Resource Name (ARN).
- D. Create an origin access identity (OAI). Assign the OAI to the CloudFront distribution. Configure the S3 bucket permissions so that only the OAI has read permission.

【參考答案】D

【題號】question 132

【題目】

A company's website provides users with downloadable historical performance reports. The website needs a solution that will scale to meet the company's website demands globally. The solution should be cost-effective, limit the provisioning of infrastructure resources, and provide the fastest possible response time.

Which combination should a solutions architect recommend to meet these requirements?

一家公司的網站為用戶提供可下載的歷史績效報告。該網站需要一個能夠擴展的解決方案，以滿足公司在全球範圍內的網站需求。此解決方案應該具有成本效益，限制基礎設施資源的配置，並提供盡可能最快的回應時間。

解決方案架構師應該推薦哪一種組合來滿足這些要求？

【選項】

- A.Amazon CloudFront and Amazon S3
- B.AWS Lambda and Amazon DynamoDB
- C.Application Load Balancer with Amazon EC2 Auto Scaling
- D.Amazon Route 53 with internal Application Load Balancers

【參考答案】A

【題號】question 133

【題目】

A company runs an Oracle database on premises. As part of the company's migration to AWS, the company wants to upgrade the database to the most recent available version. The company also wants to set up disaster recovery (DR) for the database. The company needs to minimize the operational overhead for normal operations and DR setup. The company also needs to maintain access to the database's underlying operating system.

Which solution will meet these requirements?

一家公司在本地運行 Oracle 資料庫。作為公司遷移到 AWS 的一部分，該公司希望將資料庫升級到最新的可用版本。該公司還希望為資料庫設定災難復原 (DR)。公司需要最大限度地減少正常營運和災難復原設定的營運開銷。該公司還需要維護對資料庫底層作業系統的存取。

哪種解決方案可以滿足這些要求？

【選項】

A.Migrate the Oracle database to an Amazon EC2 instance. Set up database replication to a different AWS Region.

B.Migrate the Oracle database to Amazon RDS for Oracle. Activate Cross-Region automated backups to replicate the snapshots to another AWS Region.

C.Migrate the Oracle database to Amazon RDS Custom for Oracle. Create a read replica for the database in another AWS Region.

D.Migrate the Oracle database to Amazon RDS for Oracle. Create a standby database in another Availability Zone.

【參考答案】C

【題號】question 134

【題目】

A company wants to move its application to a serverless solution. The serverless solution needs to analyze existing and new data by using SL. The company stores the data in an Amazon S3 bucket. The data requires encryption and must be replicated to a different AWS Region.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司希望將其應用程式遷移到無伺服器解決方案。無伺服器解決方案需要使用 SL 來分析現有資料和新資料。該公司將資料儲存在 Amazon S3 儲存桶中。資料需要加密並且必須複製到不同的 AWS 區域。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Create a new S3 bucket. Load the data into the new S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Use Amazon Athena to query the data.

B.Create a new S3 bucket. Load the data into the new S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Use Amazon RDS to query the data.

C.Load the data into the existing S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use Amazon Athena to query the data.

D.Load the data into the existing S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use Amazon RDS to query the data.

【參考答案】A

【題號】question 135

【題目】

A company runs workloads on AWS. The company needs to connect to a service from an external provider. The service is hosted in the provider's VPC. According to the company's security team, the connectivity must be private and must be restricted to the target service. The connection must be initiated only from the company's VPC.

Which solution will mast these requirements?

一家公司在 AWS 上運作工作負載。該公司需要連接到外部提供者的服務。該服務託管在提供者的 VPC 中。據該公司的安全團隊稱，連接必須是私有的，並且必須僅限於目標服務。連接必須僅從公司的 VPC 發起。

哪種解決方案能夠滿足這些要求？

【選項】

A.Create a VPC peering connection between the company's VPC and the provider's VPC. Update the route table to connect to the target service.

B.Ask the provider to create a virtual private gateway in its VPC. Use AWS PrivateLink to connect to the target service.

C.Create a NAT gateway in a public subnet of the company's VPC. Update the route table to connect to the target service.

D.Ask the provider to create a VPC endpoint for the target service. Use AWS PrivateLink to connect to the target service.

【參考答案】D

【題號】question 136

【題目】

A company is migrating its on-premises PostgreSQL database to Amazon Aurora PostgreSQL. The on-premises database must remain online and accessible during the migration. The Aurora database must remain synchronized with the on-premises database.

Which combination of actions must a solutions architect take to meet these requirements? (Choose two.)

一家公司正在將其本地 PostgreSQL 資料庫遷移到 Amazon Aurora PostgreSQL。本地資料庫必須在遷移期間保持線上且可存取。Aurora 資料庫必須與本機資料庫保持同步。

解決方案架構師必須採取哪些操作組合才能滿足這些要求？（選兩個。）

【選項】

- A.Create an ongoing replication task.
- B.Create a database backup of the on-premises database.
- C.Create an AWS Database Migration Service (AWS DMS) replication server.
- D.Convert the database schema by using the AWS Schema Conversion Tool (AWS SCT).
- E.Create an Amazon EventBridge (Amazon CloudWatch Events) rule to monitor the database synchronization.

【參考答案】AC

【題號】question 137

【題目】

A company uses AWS Organizations to create dedicated AWS accounts for each business unit to manage each business unit's account independently upon request. The root email recipient missed a notification that was sent to the root user email address of one account. The company wants to ensure that all future notifications are not missed. Future notifications must be limited to account administrators.

Which solution will meet these requirements?

一家公司使用 AWS Organizations 為每個業務部門建立專用 AWS 帳戶，以便根據請求獨立管理每個業務部門的帳戶。根電子郵件收件者錯過了發送到一個帳戶的根用戶電子郵件地址的通知。該公司希望確保不會錯過所有未來的通知。未來的通知必須僅限於帳戶管理員。

哪種解決方案可以滿足這些要求？

【選項】

- A.Configure the company's email server to forward notification email messages that are sent to the AWS account root user email address to all users in the organization.
- B.Configure all AWS account root user email addresses as distribution lists that go to a few administrators who can respond to alerts. Configure AWS account alternate contacts in the AWS Organizations console or programmatically.

C. Configure all AWS account root user email messages to be sent to one administrator who is responsible for monitoring alerts and forwarding those alerts to the appropriate groups.

D. Configure all existing AWS accounts and all newly created accounts to use the same root user email address. Configure AWS account alternate contacts in the AWS Organizations console or programmatically.

【參考答案】B

【題號】question 138

【題目】

A company runs its ecommerce application on AWS. Every new order is published as a message in a RabbitMQ queue that runs on an Amazon EC2 instance in a single Availability Zone. These messages are processed by a different application that runs on a separate EC2 instance. This application stores the details in a PostgreSQL database on another EC2 instance. All the EC2 instances are in the same Availability Zone.

The company needs to redesign its architecture to provide the highest availability with the least operational overhead.

What should a solutions architect do to meet these requirements?

一家公司在 AWS 上運行其電子商務應用程式。每個新訂單都會作為訊息發佈在 RabbitMQ 佇列中，該佇列在單一可用區的 Amazon EC2 執行個體上執行。這些訊息由在單獨的 EC2 執行個體上執行的不同應用程式處理。該應用程式將詳細資訊儲存在另一個 EC2 執行個體上的 PostgreSQL 資料庫中。所有 EC2 執行個體都位於同一可用區。

該公司需要重新設計其架構，以最少的營運開銷提供最高的可用性。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Migrate the queue to a redundant pair (active/standby) of RabbitMQ instances on Amazon MQ. Create a Multi-AZ Auto Scaling group for EC2 instances that host the application. Create another Multi-AZ Auto Scaling group for EC2 instances that host the PostgreSQL database.

B.Migrate the queue to a redundant pair (active/standby) of RabbitMQ instances on Amazon MQ. Create a Multi-AZ Auto Scaling group for EC2 instances that host the application. Migrate the database to run on a Multi-AZ deployment of Amazon RDS for PostgreSQL.

C.Create a Multi-AZ Auto Scaling group for EC2 instances that host the RabbitMQ queue. Create another Multi-AZ Auto Scaling group for EC2 instances that host the application. Migrate the database to run on a Multi-AZ deployment of Amazon RDS for PostgreSQL.

D.Create a Multi-AZ Auto Scaling group for EC2 instances that host the RabbitMQ queue. Create another Multi-AZ Auto Scaling group for EC2 instances that host the application. Create a third Multi-AZ Auto Scaling group for EC2 instances that host the PostgreSQL database

【參考答案】B

【題號】question 139

【題目】

A reporting team receives files each day in an Amazon S3 bucket. The reporting team manually reviews and copies the files from this initial S3 bucket to an analysis S3 bucket each day at the same time to use with Amazon QuickSight. Additional teams are starting to send more files in larger sizes to the initial S3 bucket.

The reporting team wants to move the files automatically analysis S3 bucket as the files enter the initial S3 bucket. The reporting team also wants to use AWS Lambda functions to run pattern-matching code on the copied data. In addition, the reporting team wants to send the data files to a pipeline in Amazon SageMaker Pipelines.

What should a solutions architect do to meet these requirements with the LEAST operational overhead?

報告團隊每天都會收到 Amazon S3 儲存桶中的文件。報告團隊每天同時手動查看此初始 S3 儲存桶中的檔案並將其複製到分析 S3 儲存桶，以與 Amazon QuickSight 一起使用。更多團隊開始將更多更大尺寸的檔案傳送到初始 S3 儲存桶。

報告團隊希望在文件進入初始 S3 儲存桶時自動移動文件分析 S3 儲存桶。報告團隊也希望使用 AWS Lambda 函數對複製的資料運行模式來匹配程式碼。此外，報告團隊希望將資料檔案傳送到 Amazon SageMaker Pipelines 中的管道。

解決方案架構師應該如何做才能以最少的營運開銷來滿足這些要求？

【選項】

A.Create a Lambda function to copy the files to the analysis S3 bucket. Create an S3 event notification for the analysis S3 bucket. Configure Lambda and SageMaker Pipelines as destinations of the event notification. Configure s3:ObjectCreated:Put as the event type.

B.Create a Lambda function to copy the files to the analysis S3 bucket. Configure the analysis S3 bucket to send event notifications to Amazon EventBridge (Amazon CloudWatch Events). Configure an ObjectCreated rule in EventBridge (CloudWatch Events). Configure Lambda and SageMaker Pipelines as targets for the rule.

C.Configure S3 replication between the S3 buckets. Create an S3 event notification for the analysis S3 bucket. Configure Lambda and SageMaker Pipelines as destinations of the event notification. Configure s3:ObjectCreated:Put as the event type.

D.Configure S3 replication between the S3 buckets. Configure the analysis S3 bucket to send event notifications to Amazon EventBridge (Amazon CloudWatch Events). Configure an ObjectCreated rule in EventBridge (CloudWatch Events). Configure Lambda and SageMaker Pipelines as targets for the rule.

【參考答案】D

【題號】question 140

【題目】

A solutions architect needs to help a company optimize the cost of running an application on AWS. The application will use Amazon EC2 instances, AWS Fargate, and AWS Lambda for compute within the architecture.

The EC2 instances will run the data ingestion layer of the application. EC2 usage will be sporadic and unpredictable. Workloads that run on EC2 instances can be interrupted at any time. The application front end will run on Fargate, and Lambda will serve the API layer. The front-end utilization and API layer utilization will be predictable over the course of the next year.

Which combination of purchasing options will provide the MOST cost-effective solution for hosting this application? (Choose two.)

解決方案架構師需要協助公司優化在 AWS 上執行應用程式的成本。該應用程式將使用 Amazon EC2 執行個體、AWS Fargate 和 AWS Lambda 在架構內進行運算。

EC2 執行個體將執行應用程式的資料擷取層。EC2 的使用將是零星且不可預測的。在 EC2 執行個體上執行的工作負載可以隨時中斷。應用程式前端將在 Fargate 上運行，Lambda 將服務 API 層。明年的前端利用率和 API 層利用率將是可預測的。

哪種購買選項組合將為託管此應用程式提供最具成本效益的解決方案？（選兩個。）

【選項】

- A. Use Spot Instances for the data ingestion layer
- B. Use On-Demand Instances for the data ingestion layer
- C. Purchase a 1-year Compute Savings Plan for the front end and API layer.
- D. Purchase 1-year All Upfront Reserved instances for the data ingestion layer.
- E. Purchase a 1-year EC2 instance Savings Plan for the front end and API layer.

【參考答案】AC

【題號】question 141

【題目】

A company runs a web-based portal that provides users with global breaking news, local alerts, and weather updates. The portal delivers each user a personalized view by using mixture of static and dynamic content. Content is served over HTTPS through an API server running on an Amazon EC2 instance behind an Application Load Balancer (ALB). The company wants the portal to provide this content to its users across the world as quickly as possible.

How should a solutions architect design the application to ensure the LEAST amount of latency for all users?

一家公司經營一個基於網路的入口網站，為用戶提供全球突發新聞、本地警報和天氣更新。此入口網站透過混合使用靜態和動態內容為每個使用者提供個人化視圖。內容透過在應用程式負載平衡器 (ALB) 後面的 Amazon EC2 執行個體上執行的 API 伺服器透過 HTTPS 提供。該公司希望該入口網站盡快向世界各地的用戶提供此內容。

解決方案架構師應如何設計應用程式以確保所有使用者的延遲最少？

【選項】

A. Deploy the application stack in a single AWS Region. Use Amazon CloudFront to serve all static and dynamic content by specifying the ALB as an origin.

B. Deploy the application stack in two AWS Regions. Use an Amazon Route 53 latency routing policy to serve all content from the ALB in the closest Region.

C. Deploy the application stack in a single AWS Region. Use Amazon CloudFront to serve the static content. Serve the dynamic content directly from the ALB.

D. Deploy the application stack in two AWS Regions. Use an Amazon Route 53 geolocation routing policy to serve all content from the ALB in the closest Region.

【參考答案】A

【題號】question 142

【題目】

A gaming company is designing a highly available architecture. The application runs on a modified Linux kernel and supports only UDP-based traffic. The company needs the front-end tier to provide the best possible user experience. That tier must have low latency, route traffic to the nearest edge location, and provide static IP addresses for entry into the application endpoints.

What should a solutions architect do to meet these requirements?

一家遊戲公司正在設計一個高度可用的架構。該應用程式在修改後的 Linux 核心上運行，並且僅支援基於 UDP 的流量。該公司需要前端層來提供最佳的用戶體驗。此層必須具有低延遲、將流量路由到最近的邊緣位置，並提供用於進入應用程式端點的靜態 IP 位址。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Configure Amazon Route 53 to forward requests to an Application Load Balancer. Use AWS Lambda for the application in AWS Application Auto Scaling.

B. Configure Amazon CloudFront to forward requests to a Network Load Balancer. Use AWS Lambda for the application in an AWS Application Auto Scaling group.

C. Configure AWS Global Accelerator to forward requests to a Network Load Balancer. Use Amazon EC2 instances for the application in an EC2 Auto Scaling group.

D. Configure Amazon API Gateway to forward requests to an Application Load Balancer. Use Amazon EC2 instances for the application in an EC2 Auto Scaling group.

【參考答案】C

【題號】question 143

【題目】

A company wants to migrate its existing on-premises monolithic application to AWS. The company wants to keep as much of the front-end code and the backend code as possible. However, the company wants to break the application into smaller applications. A different team will manage each application. The company needs a highly scalable solution that minimizes operational overhead.

Which solution will meet these requirements?

一家公司希望將其現有的本地整體應用程式遷移到 AWS。該公司希望保留盡可能多的前端代碼和後端代碼。然而，該公司希望將應用程式分解為更小的應用程式。不同的團隊將管理每個應用程式。該公司需要一個高度可擴展的解決方案，以最大限度地減少營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A. Host the application on AWS Lambda. Integrate the application with Amazon API Gateway.

B. Host the application with AWS Amplify. Connect the application to an Amazon API Gateway API that is integrated with AWS Lambda.

C.Host the application on Amazon EC2 instances. Set up an Application Load Balancer with EC2 instances in an Auto Scaling group as targets.

D.Host the application on Amazon Elastic Container Service (Amazon ECS). Set up an Application Load Balancer with Amazon ECS as the target.

【參考答案】D

【題號】question 144

【題目】

A company recently started using Amazon Aurora as the data store for its global ecommerce application. When large reports are run, developers report that the ecommerce application is performing poorly. After reviewing metrics in Amazon CloudWatch, a solutions architect finds that the ReadIOPS and CPUUtilization metrics are spiking when monthly reports run.

What is the MOST cost-effective solution?

一家公司最近開始使用 Amazon Aurora 作為其全球電子商務應用程式的資料儲存。執行大型報告時，開發人員報告電子商務應用程式效能不佳。在檢查 Amazon CloudWatch 中的指標後，解決方案架構師發現 ReadIOPS 和 CPUUtilization 指標在每月報告執行時出現高峰。

最具成本效益的解決方案是什麼？

【選項】

A.Migrate the monthly reporting to Amazon Redshift.

B.Migrate the monthly reporting to an Aurora Replica.

C.Migrate the Aurora database to a larger instance class.

D.Increase the Provisioned IOPS on the Aurora instance.

【參考答案】B

【題號】question 145

【題目】

A company hosts a website analytics application on a single Amazon EC2 On-Demand Instance. The analytics software is written in PHP and uses a MySQL database. The analytics software, the web server that provides PHP, and the database server are all hosted on the EC2 instance. The application is showing signs of performance degradation during busy times and is presenting 5xx errors. The company needs to make the application scale seamlessly.

Which solution will meet these requirements MOST cost-effectively?

一家公司在單一 Amazon EC2 隨選執行個體上託管網站分析應用程式。此分析軟體是用 PHP 編寫的，並使用 MySQL 資料庫。分析軟體、提供 PHP 的 Web 伺服器以及資料庫伺服器都託管在 EC2 執行個體上。該應用程式在繁忙時間顯示出效能下降的跡象，並出現 5xx 錯誤。公司需要使應用程式無縫擴展。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A.Migrate the database to an Amazon RDS for MySQL DB instance. Create an AMI of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use an Application Load Balancer to distribute the load to each EC2 instance.

B.Migrate the database to an Amazon RDS for MySQL DB instance. Create an AMI of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use Amazon Route 53 weighted routing to distribute the load across the two EC2 instances.

C.Migrate the database to an Amazon Aurora MySQL DB instance. Create an AWS Lambda function to stop the EC2 instance and change the instance type. Create an Amazon CloudWatch alarm to invoke the Lambda function when CPU utilization surpasses 75%.

D.Migrate the database to an Amazon Aurora MySQL DB instance. Create an AMI of the web application. Apply the AMI to a launch template. Create an Auto Scaling group with the launch template. Configure the launch template to use a Spot Fleet. Attach an Application Load Balancer to the Auto Scaling group.

【參考答案】D

【題號】question 146

【題目】

A company runs a stateless web application in production on a group of Amazon EC2 On-Demand Instances behind an Application Load Balancer. The application experiences heavy usage during an 8-hour period each business day. Application usage is moderate and steady overnight. Application usage is low during weekends.

The company wants to minimize its EC2 costs without affecting the availability of the application.

Which solution will meet these requirements?

一家公司在應用程式負載平衡器後面的一組 Amazon EC2 隨選執行個體上執行生產中的無狀態 Web 應用程式。該應用程式在每個工作日的 8 小時內都會經歷大量使用。夜間應用程式使用情況適度且穩定。週末應用程式使用率較低。

該公司希望在不影響應用程式可用性的情況下最大限度地降低 EC2 成本。

哪種解決方案可以滿足這些要求？

【選項】

A. Use Spot Instances for the entire workload.

B. Use Reserved Instances for the baseline level of usage. Use Spot instances for any additional capacity that the application needs.

C. Use On-Demand Instances for the baseline level of usage. Use Spot Instances for any additional capacity that the application needs.

D. Use Dedicated Instances for the baseline level of usage. Use On-Demand Instances for any additional capacity that the application needs.

【參考答案】B

【題號】question 147

【題目】

A company needs to retain application log files for a critical application for 10 years. The application team regularly accesses logs from the past month for troubleshooting, but logs older than 1 month are rarely accessed. The application generates more than 10 TB of logs per month.

Which storage option meets these requirements MOST cost-effectively?

公司需要將關鍵應用程式的應用程式日誌檔案保留 10 年。應用程式團隊定期存取過去一個月的日誌以進行故障排除，但很少存取超過 1 個月的日誌。該應用程式每月產生超過 10 TB 的日誌。

哪種儲存選項最經濟有效率地滿足這些要求？

【選項】

A.Store the logs in Amazon S3. Use AWS Backup to move logs more than 1 month old to S3 Glacier Deep Archive.

B.Store the logs in Amazon S3. Use S3 Lifecycle policies to move logs more than 1 month old to S3 Glacier Deep Archive.

C.Store the logs in Amazon CloudWatch Logs. Use AWS Backup to move logs more than 1 month old to S3 Glacier Deep Archive.

D.Store the logs in Amazon CloudWatch Logs. Use Amazon S3 Lifecycle policies to move logs more than 1 month old to S3 Glacier Deep Archive.

【參考答案】B

【題號】question 148

【題目】

A company has a data ingestion workflow that includes the following components:

An Amazon Simple Notification Service (Amazon SNS) topic that receives notifications about new data deliveries

An AWS Lambda function that processes and stores the data

The ingestion workflow occasionally fails because of network connectivity issues. When failure occurs, the corresponding data is not ingested unless the company manually reruns the job.

What should a solutions architect do to ensure that all notifications are eventually processed?

一家公司的資料攝取工作流程包括以下組件：

接收有關新資料傳輸的通知的 Amazon Simple Notification Service (Amazon SNS) 主題
處理和儲存資料的 AWS Lambda 函數

由於網路連線問題，攝取工作流程有時會失敗。當發生故障時，除非公司手動重新執行作業，否則不會攝取對應的資料。

解決方案架構師應該如何確保所有通知最終得到處理？

【選項】

A. Configure the Lambda function for deployment across multiple Availability Zones.

B. Modify the Lambda function's configuration to increase the CPU and memory allocations for the function.

C. Configure the SNS topic's retry strategy to increase both the number of retries and the wait time between retries.

D. Configure an Amazon Simple Queue Service (Amazon SQS) queue as the on-failure destination. Modify the Lambda function to process messages in the queue.

【參考答案】D

【題號】question 149

【題目】

A company has a service that produces event data. The company wants to use AWS to process the event data as it is received. The data is written in a specific order that must be

maintained throughout processing. The company wants to implement a solution that minimizes operational overhead.

How should a solutions architect accomplish this?

一家公司提供產生事件資料的服務。該公司希望使用 AWS 處理收到的事件資料。資料以特定順序寫入，在整個處理過程中必須保持該順序。該公司希望實施一種能夠最大限度地減少營運開銷的解決方案。

解決方案架構師應該如何實現這個目標？

【選項】

A.Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue to hold messages. Set up an AWS Lambda function to process messages from the queue.

B.Create an Amazon Simple Notification Service (Amazon SNS) topic to deliver notifications containing payloads to process. Configure an AWS Lambda function as a subscriber.

C.Create an Amazon Simple Queue Service (Amazon SQS) standard queue to hold messages. Set up an AWS Lambda function to process messages from the queue independently.

D.Create an Amazon Simple Notification Service (Amazon SNS) topic to deliver notifications containing payloads to process. Configure an Amazon Simple Queue Service (Amazon SQS) queue as a subscriber.

【參考答案】A

【題號】question 150

【題目】

A company is migrating an application from on-premises servers to Amazon EC2 instances. As part of the migration design requirements, a solutions architect must implement infrastructure metric alarms. The company does not need to take action if CPU utilization increases to more than 50% for a short burst of time. However, if the CPU utilization increases to more than 50% and read IOPS on the disk are high at the same

time, the company needs to act as soon as possible. The solutions architect also must reduce false alarms.

What should the solutions architect do to meet these requirements?

一家公司正在將應用程式從本機伺服器遷移到 Amazon EC2 執行個體。作為遷移設計要求的一部分，解決方案架構師必須實現基礎設施指標警報。如果CPU利用率在短時間內增加到50%以上，公司不需要採取措施。但如果CPU利用率上升到50%以上，同時磁碟讀取IOPS很高，公司就需要盡快採取行動。解決方案架構師也必須減少誤報。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Create Amazon CloudWatch composite alarms where possible.
- B.Create Amazon CloudWatch dashboards to visualize the metrics and react to issues quickly.
- C.Create Amazon CloudWatch Synthetics canaries to monitor the application and raise an alarm.
- D.Create single Amazon CloudWatch metric alarms with multiple metric thresholds where possible.

【參考答案】A

【題號】question 151

【題目】

A company wants to migrate its on-premises data center to AWS. According to the company's compliance requirements, the company can use only the ap-northeast-3 Region. Company administrators are not permitted to connect VPCs to the internet.

Which solutions will meet these requirements? (Choose two.)

一家公司希望將其本地資料中心遷移到 AWS。根據公司合規要求，公司只能使用 ap-northeast-3 Region。公司管理員無權將VPC連接到網路。

哪些解決方案可以滿足這些要求？（選兩個。）

【選項】

- A. Use AWS Control Tower to implement data residency guardrails to deny internet access and deny access to all AWS Regions except ap-northeast-3.
- B. Use rules in AWS WAF to prevent internet access. Deny access to all AWS Regions except ap-northeast-3 in the AWS account settings.
- C. Use AWS Organizations to configure service control policies (SCPS) that prevent VPCs from gaining internet access. Deny access to all AWS Regions except ap-northeast-3.
- D. Create an outbound rule for the network ACL in each VPC to deny all traffic from 0.0.0.0/0. Create an IAM policy for each user to prevent the use of any AWS Region other than ap-northeast-3.
- E. Use AWS Config to activate managed rules to detect and alert for internet gateways and to detect and alert for new resources deployed outside of ap-northeast-3.

【參考答案】AC

【題號】question 152

【題目】

A company uses a three-tier web application to provide training to new employees. The application is accessed for only 12 hours every day. The company is using an Amazon RDS for MySQL DB instance to store information and wants to minimize costs.

What should a solutions architect do to meet these requirements?

一家公司使用三層 Web 應用程式為新員工提供培訓。每天僅訪問該應用程式 12 小時。該公司正在使用 Amazon RDS for MySQL 資料庫執行個體來儲存訊息，並希望最大限度地降低成本。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Configure an IAM policy for AWS Systems Manager Session Manager. Create an IAM role for the policy. Update the trust relationship of the role. Set up automatic start and stop for the DB instance.

B. Create an Amazon ElastiCache for Redis cache cluster that gives users the ability to access the data from the cache when the DB instance is stopped. Invalidate the cache after the DB instance is started.

C. Launch an Amazon EC2 instance. Create an IAM role that grants access to Amazon RDS. Attach the role to the EC2 instance. Configure a cron job to start and stop the EC2 instance on the desired schedule.

D. Create AWS Lambda functions to start and stop the DB instance. Create Amazon EventBridge (Amazon CloudWatch Events) scheduled rules to invoke the Lambda functions. Configure the Lambda functions as event targets for the rules.

【參考答案】D

【題號】question 153

【題目】

A company sells ringtones created from clips of popular songs. The files containing the ringtones are stored in Amazon S3 Standard and are at least 128 KB in size. The company has millions of files, but downloads are infrequent for ringtones older than 90 days. The company needs to save money on storage while keeping the most accessed files readily available for its users.

Which action should the company take to meet these requirements MOST cost-effectively?

一家公司銷售根據流行歌曲片段製作的鈴聲。包含鈴聲的檔案儲存在 Amazon S3 Standard 中，大小至少為 128 KB。該公司擁有數百萬個文件，但超過 90 天的鈴聲下載很少。該公司需要節省儲存成本，同時讓用戶隨時可以存取最常存取的文件。

公司應該採取哪些行動來以最具成本效益的方式滿足這些要求？

【選項】

- A. Configure S3 Standard-Infrequent Access (S3 Standard-IA) storage for the initial storage tier of the objects.
- B. Move the files to S3 Intelligent-Tiering and configure it to move objects to a less expensive storage tier after 90 days.
- C. Configure S3 inventory to manage objects and move them to S3 Standard-Infrequent Access (S3 Standard-1A) after 90 days.
- D. Implement an S3 Lifecycle policy that moves the objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-1A) after 90 days.

【參考答案】D

【題號】question 154

【題目】

A company needs to save the results from a medical trial to an Amazon S3 repository. The repository must allow a few scientists to add new files and must restrict all other users to read-only access. No users can have the ability to modify or delete any files in the repository. The company must keep every file in the repository for a minimum of 1 year after its creation date.

Which solution will meet these requirements?

公司需要將醫學試驗的結果儲存到 Amazon S3 儲存庫。儲存庫必須允許少數科學家添加新文件，並且必須限制所有其他使用者的唯讀存取權限。任何使用者都無法修改或刪除儲存庫中的任何檔案。公司必須將每個文件在創建之日起至少保留 1 年。

哪種解決方案可以滿足這些要求？

【選項】

- A. Use S3 Object Lock in governance mode with a legal hold of 1 year.
- B. Use S3 Object Lock in compliance mode with a retention period of 365 days.
- C. Use an IAM role to restrict all users from deleting or changing objects in the S3 bucket. Use an S3 bucket policy to only allow the IAM role.

D.Configure the S3 bucket to invoke an AWS Lambda function every time an object is added. Configure the function to track the hash of the saved object so that modified objects can be marked accordingly.

【參考答案】B

【題號】question 155

【題目】

A large media company hosts a web application on AWS. The company wants to start caching confidential media files so that users around the world will have reliable access to the files. The content is stored in Amazon S3 buckets. The company must deliver the content quickly, regardless of where the requests originate geographically.

Which solution will meet these requirements?

一家大型媒體公司在 AWS 上託管 Web 應用程式。該公司希望開始快取機密媒體文件，以便世界各地的用戶能夠可靠地存取這些文件。內容儲存在 Amazon S3 儲存桶中。無論請求來自何處，公司都必須快速交付內容。

哪種解決方案可以滿足這些要求？

【選項】

- A.Use AWS DataSync to connect the S3 buckets to the web application.
- B.Deploy AWS Global Accelerator to connect the S3 buckets to the web application.
- C.Deploy Amazon CloudFront to connect the S3 buckets to CloudFront edge servers.
- D.Use Amazon Simple Queue Service (Amazon SQS) to connect the S3 buckets to the web application.

【參考答案】C

【題號】question 156

【題目】

A company produces batch data that comes from different databases. The company also produces live stream data from network sensors and application APIs. The company needs to consolidate all the data into one place for business analytics. The company needs to process the incoming data and then stage the data in different Amazon S3 buckets. Teams will later run one-time queries and import the data into a business intelligence tool to show key performance indicators (KPIs).

Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

一家公司產生來自不同資料庫的大量資料。該公司還透過網路感測器和應用程式 API 產生即時串流數據。該公司需要將所有數據整合到一處進行業務分析。該公司需要處理傳入的數據，然後將數據暫存在不同的 Amazon S3 儲存桶中。團隊稍後將執行一次性查詢並將資料匯入商業智慧工具以顯示關鍵績效指標 (KPI)。

哪種步驟組合能夠以最少的營運開銷滿足這些要求？（選兩個。）

【選項】

A. Use Amazon Athena for one-time queries. Use Amazon QuickSight to create dashboards for KPIs.

B. Use Amazon Kinesis Data Analytics for one-time queries. Use Amazon QuickSight to create dashboards for KPIs.

C. Create custom AWS Lambda functions to move the individual records from the databases to an Amazon Redshift cluster.

D. Use an AWS Glue extract, transform, and load (ETL) job to convert the data into JSON format. Load the data into multiple Amazon OpenSearch Service (Amazon Elasticsearch Service) clusters.

E. Use blueprints in AWS Lake Formation to identify the data that can be ingested into a data lake. Use AWS Glue to crawl the source, extract the data, and load the data into Amazon S3 in Apache Parquet format.

【參考答案】AE

【題號】question 157

【題目】

A company stores data in an Amazon Aurora PostgreSQL DB cluster. The company must store all the data for 5 years and must delete all the data after 5 years. The company also must indefinitely keep audit logs of actions that are performed within the database. Currently, the company has automated backups configured for Aurora.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

一家公司將資料儲存在 Amazon Aurora PostgreSQL 資料庫叢集中。公司必須將所有資料保存5年，並在5年後刪除所有資料。公司還必須無限期保留資料庫中執行的操作的審核日誌。目前，該公司已為 Aurora 配置了自動備份。

解決方案架構師應該採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

- A.Take a manual snapshot of the DB cluster.
- B.Create a lifecycle policy for the automated backups.
- C.Configure automated backup retention for 5 years.
- D.Configure an Amazon CloudWatch Logs export for the DB cluster.
- E.Use AWS Backup to take the backups and to keep the backups for 5 years.

【參考答案】DE

【題號】question 158

【題目】

A solutions architect is optimizing a website for an upcoming musical event. Videos of the performances will be streamed in real time and then will be available on demand. The event is expected to attract a global online audience.

Which service will improve the performance of both the real-time and on-demand streaming?

解決方案架構師正在為即將舉行的音樂活動優化網站。表演視訊將即時傳輸，然後按需提供。該活動預計將吸引全球線上觀眾。

哪種服務可以提高即時串流媒體和點播串流媒體的效能？

【選項】

- A.Amazon CloudFront
- B.AWS Global Accelerator
- C.Amazon Route 53
- D.Amazon S3 Transfer Acceleration

【參考答案】A

【題號】question 159

【題目】

A company is running a publicly accessible serverless application that uses Amazon API Gateway and AWS Lambda. The application's traffic recently spiked due to fraudulent requests from botnets.

Which steps should a solutions architect take to block requests from unauthorized users? (Choose two.)

一家公司正在運行一個使用 Amazon API Gateway 和 AWS Lambda 的可公開存取的無伺服器應用程式。由於來自殭屍網路的詐騙請求，該應用程式的流量最近激增。

解決方案架構師應採取哪些步驟來阻止未經授權使用者的請求？（選兩個。）

【選項】

- A.Create a usage plan with an API key that is shared with genuine users only.

- B.Integrate logic within the Lambda function to ignore the requests from fraudulent IP addresses.
- C.Implement an AWS WAF rule to target malicious requests and trigger actions to filter them out.
- D.Convert the existing public API to a private API. Update the DNS records to redirect users to the new API endpoint.
- E.Create an IAM role for each user attempting to access the API. A user will assume the role when making the API call.

【參考答案】AC

【題號】question 160

【題目】

An ecommerce company hosts its analytics application in the AWS Cloud. The application generates about 300 MB of data each month. The data is stored in JSON format. The company is evaluating a disaster recovery solution to back up the data. The data must be accessible in milliseconds if it is needed, and the data must be kept for 30 days.

Which solution meets these requirements MOST cost-effectively?

一家電子商務公司在 AWS 雲端中託管其分析應用程式。該應用程式每月產生約 300 MB 的數據。資料以 JSON 格式儲存。該公司正在評估備份資料的災難復原解決方案。如果需要，資料必須可以在毫秒內訪問，並且資料必須保留 30 天。

哪種解決方案最經濟有效率地滿足這些要求？

【選項】

- A.Amazon OpenSearch Service (Amazon Elasticsearch Service)
- B.Amazon S3 Glacier
- C.Amazon S3 Standard
- D.Amazon RDS for PostgreSQL

【參考答案】C

【題號】question 161

【題目】

A company has a small Python application that processes JSON documents and outputs the results to an on-premises SQL database. The application runs thousands of times each day. The company wants to move the application to the AWS Cloud. The company needs a highly available solution that maximizes scalability and minimizes operational overhead.

Which solution will meet these requirements?

一家公司有一個小型 Python 應用程式，用於處理 JSON 文件並將結果輸出到本地 SQL 資料庫。該應用程式每天運行數千次。該公司希望將應用程式遷移到 AWS 雲端。該公司需要一個高度可用的解決方案，最大限度地提高可擴展性並最大限度地減少營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A.Place the JSON documents in an Amazon S3 bucket. Run the Python code on multiple Amazon EC2 instances to process the documents. Store the results in an Amazon Aurora DB cluster.

B.Place the JSON documents in an Amazon S3 bucket. Create an AWS Lambda function that runs the Python code to process the documents as they arrive in the S3 bucket. Store the results in an Amazon Aurora DB cluster.

C.Place the JSON documents in an Amazon Elastic Block Store (Amazon EBS) volume. Use the EBS Multi-Attach feature to attach the volume to multiple Amazon EC2 instances. Run the Python code on the EC2 instances to process the documents. Store the results on an Amazon RDS DB instance.

D.Place the JSON documents in an Amazon Simple Queue Service (Amazon SQS) queue as messages. Deploy the Python code as a container on an Amazon Elastic Container Service (Amazon ECS) cluster that is configured with the Amazon EC2 launch type. Use the container to process the SQS messages. Store the results on an Amazon RDS DB instance.

【參考答案】B

【題號】question 162

【題目】

A company wants to use high performance computing (HPC) infrastructure on AWS for financial risk modeling. The company's HPC workloads run on Linux. Each HPC workflow runs on hundreds of Amazon EC2 Spot Instances, is short-lived, and generates thousands of output files that are ultimately stored in persistent storage for analytics and long-term future use.

The company seeks a cloud storage solution that permits the copying of on-premises data to long-term persistent storage to make data available for processing by all EC2 instances. The solution should also be a high performance file system that is integrated with persistent storage to read and write datasets and output files.

Which combination of AWS services meets these requirements?

一家公司希望使用 AWS 上的高效能運算 (HPC) 基礎架構進行財務風險建模。該公司的 HPC 工作負載在 Linux 上運作。每個 HPC 工作流程在數百個 Amazon EC2 Spot 執行個體上運行，生命週期短暫，並產生數千個輸出文件，這些文件最終儲存在持久性儲存中以供分析和未來長期使用。

該公司尋求一種雲端儲存解決方案，允許將本地資料複製到長期持久存儲，以使資料可供所有 EC2 執行個體處理。該解決方案還應該是一個與持久性儲存整合的高效能檔案系統，用於讀寫資料集和輸出檔案。

哪種 AWS 服務組合可以滿足這些要求？

【選項】

A.Amazon FSx for Lustre integrated with Amazon S3

B.Amazon FSx for Windows File Server integrated with Amazon S3

C.Amazon S3 Glacier integrated with Amazon Elastic Block Store (Amazon EBS)

D.Amazon S3 bucket with a VPC endpoint integrated with an Amazon Elastic Block Store (Amazon EBS) General Purpose SSD (gp2) volume

【參考答案】A

【題號】question 163

【題目】

A company is building a containerized application on premises and decides to move the application to AWS. The application will have thousands of users soon after it is deployed. The company is unsure how to manage the deployment of containers at scale. The company needs to deploy the containerized application in a highly available architecture that minimizes operational overhead.

Which solution will meet these requirements?

一家公司正在本地建立容器化應用程式，並決定將該應用程式遷移到 AWS。該應用程式部署後很快就會擁有數千名用戶。該公司不確定如何管理大規模容器部署。該公司需要在高度可用的架構中部署容器化應用程式，以最大限度地減少營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A.Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the AWS Fargate launch type to run the containers. Use target tracking to scale automatically based on demand.

B.Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the Amazon EC2 launch type to run the containers. Use target tracking to scale automatically based on demand.

C.Store container images in a repository that runs on an Amazon EC2 instance. Run the containers on EC2 instances that are spread across multiple Availability Zones. Monitor the average CPU utilization in Amazon CloudWatch. Launch new EC2 instances as needed.

D.Create an Amazon EC2 Amazon Machine Image (AMI) that contains the container image. Launch EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon CloudWatch alarm to scale out EC2 instances when the average CPU utilization threshold is breached.

【參考答案】A

【題號】question 164

【題目】

A company has two applications: a sender application that sends messages with payloads to be processed and a processing application intended to receive the messages with payloads. The company wants to implement an AWS service to handle messages between the two applications. The sender application can send about 1,000 messages each hour. The messages may take up to 2 days to be processed. If the messages fail to process, they must be retained so that they do not impact the processing of any remaining messages.

Which solution meets these requirements and is the MOST operationally efficient?

該公司有兩個應用程式：發送者應用程式發送帶有要處理的有效負載的訊息，處理應用程式旨在接收帶有有效負載的訊息。該公司希望實施 AWS 服務來處理兩個應用程式之間的訊息。發送方應用程式每小時可以發送約 1,000 則訊息。處理訊息最多可能需要 2 天：如果訊息處理失敗，則必須保留它們，以免影響任何剩餘訊息的處理。

哪種解決方案滿足這些要求且營運效率最高？

【選項】

- A. Set up an Amazon EC2 instance running a Redis database. Configure both applications to use the instance. Store, process, and delete the messages, respectively.
- B. Use an Amazon Kinesis data stream to receive the messages from the sender application. Integrate the processing application with the Kinesis Client Library (KCL).
- C. Integrate the sender and processor applications with an Amazon Simple Queue Service (Amazon SQS) queue. Configure a dead-letter queue to collect the messages that failed to process.
- D. Subscribe the processing application to an Amazon Simple Notification Service (Amazon SNS) topic to receive notifications to process. Integrate the sender application to write to the SNS topic.

【參考答案】C

【題號】question 165

【題目】

A solutions architect must design a solution that uses Amazon CloudFront with an Amazon S3 origin to store a static website. The company's security policy requires that all website traffic be inspected by AWS WAF.

How should the solutions architect comply with these requirements?

解決方案架構師必須設計一個使用 Amazon CloudFront 和 Amazon S3 來源來儲存靜態網站的解決方案。該公司的安全策略要求所有網站流量均由 AWS WAF 檢查。

解決方案架構師應該如何滿足這些要求？

【選項】

A. Configure an S3 bucket policy to accept requests coming from the AWS WAF Amazon Resource Name (ARN) only.

B. Configure Amazon CloudFront to forward all incoming requests to AWS WAF before requesting content from the S3 origin.

C. Configure a security group that allows Amazon CloudFront IP addresses to access Amazon S3 only. Associate AWS WAF to CloudFront.

D. Configure Amazon CloudFront and Amazon S3 to use an origin access identity (OAI) to restrict access to the S3 bucket. Enable AWS WAF on the distribution.

【參考答案】D

【題號】question 166

【題目】

Organizers for a global event want to put daily reports online as static HTML pages. The pages are expected to generate millions of views from users around the world. The files are stored in an Amazon S3 bucket. A solutions architect has been asked to design an efficient and effective solution.

Which action should the solutions architect take to accomplish this?

全球活動的組織者希望將每日報告作為靜態 HTML 頁面放到網路上。這些頁面預計將產生來自世界各地用戶的數百萬次瀏覽。這些檔案儲存在 Amazon S3 儲存桶中。解決方案架構師被要求設計一個高效且有效的解決方案。

解決方案架構師應該採取哪些行動來實現這個目標？

【選項】

- A.Generate presigned URLs for the files.
- B.Use cross-Region replication to all Regions.
- C.Use the geoproximity feature of Amazon Route 53.
- D.Use Amazon CloudFront with the S3 bucket as its origin.

【參考答案】D

【題號】question 167

【題目】

A company runs a production application on a fleet of Amazon EC2 instances. The application reads the data from an Amazon SQS queue and processes the messages in parallel. The message volume is unpredictable and often has intermittent traffic. This application should continually process messages without any downtime.

Which solution meets these requirements MOST cost-effectively?

一家公司在一組 Amazon EC2 執行個體上執行生產應用程式。該應用程式從 Amazon SQS 佇列讀取資料並並行處理訊息。訊息量無法預測，且經常出現間歇性流量。該應用程式應該持續處理訊息而不會造成任何停機。

哪種解決方案最經濟有效率地滿足這些要求？

【選項】

- A. Use Spot Instances exclusively to handle the maximum capacity required.
- B. Use Reserved Instances exclusively to handle the maximum capacity required.
- C. Use Reserved Instances for the baseline capacity and use Spot Instances to handle additional capacity.
- D. Use Reserved Instances for the baseline capacity and use On-Demand Instances to handle additional capacity.

【參考答案】C

【題號】question 168

【題目】

A security team wants to limit access to specific services or actions in all of the team's AWS accounts. All accounts belong to a large organization in AWS Organizations. The solution must be scalable and there must be a single point where permissions can be maintained.

What should a solutions architect do to accomplish this?

安全團隊希望限制對團隊所有 AWS 帳戶中特定服務或操作的存取。所有帳戶都屬於 AWS Organizations 中的一個大型組織。該解決方案必須是可擴展的，並且必須有一個可以維護權限的點。

解決方案架構師應該做什麼來實現這個目標？

【選項】

- A. Create an ACL to provide access to the services or actions.
- B. Create a security group to allow accounts and attach it to user groups.
- C. Create cross-account roles in each account to deny access to the services or actions.

D.Create a service control policy in the root organizational unit to deny access to the services or actions.

【參考答案】D

【題號】question 169

【題目】

A company is concerned about the security of its public web application due to recent web attacks. The application uses an Application Load Balancer (ALB). A solutions architect must reduce the risk of DDoS attacks against the application.

What should the solutions architect do to meet this requirement?

由於最近的網路攻擊，一家公司擔心其公共網路應用程式的安全性。該應用程式使用應用程式負載平衡器 (ALB)。解決方案架構師必須降低應用程式遭受 DDoS 攻擊的風險。

解決方案架構師應該做什麼來滿足這個要求？

【選項】

- A.Add an Amazon Inspector agent to the ALB.
- B.Configure Amazon Macie to prevent attacks.
- C.Enable AWS Shield Advanced to prevent attacks.
- D.Configure Amazon GuardDuty to monitor the ALB.

【參考答案】C

【題號】question 170

【題目】

A company's web application is running on Amazon EC2 instances behind an Application Load Balancer. The company recently changed its policy, which now requires the application to be accessed from one specific country only.

Which configuration will meet this requirement?

一家公司的 Web 應用程式在應用程式負載平衡器後面的 Amazon EC2 執行個體上執行。該公司最近改變了政策，現在要求只能從一個特定國家/地區存取該應用程式。

哪種配置可以滿足這個要求？

【選項】

- A. Configure the security group for the EC2 instances.
- B. Configure the security group on the Application Load Balancer.
- C. Configure AWS WAF on the Application Load Balancer in a VPC.
- D. Configure the network ACL for the subnet that contains the EC2 instances.

【參考答案】C

【題號】question 171

【題目】

A company provides an API to its users that automates inquiries for tax computations based on item prices. The company experiences a larger number of inquiries during the holiday season only that cause slower response times. A solutions architect needs to design a solution that is scalable and elastic.

What should the solutions architect do to accomplish this?

一家公司向其用戶提供 API，自動查詢基於商品價格的稅金計算。該公司在假期期間遇到大量詢問，導致回應時間變慢。解決方案架構師需要設計可擴展且有彈性的解決方案。

解決方案架構師應該做什麼來實現這個目標？

【選項】

A. Provide an API hosted on an Amazon EC2 instance. The EC2 instance performs the required computations when the API request is made.

B. Design a REST API using Amazon API Gateway that accepts the item names. API Gateway passes item names to AWS Lambda for tax computations.

C. Create an Application Load Balancer that has two Amazon EC2 instances behind it. The EC2 instances will compute the tax on the received item names.

D. Design a REST API using Amazon API Gateway that connects with an API hosted on an Amazon EC2 instance. API Gateway accepts and passes the item names to the EC2 instance for tax computations.

【參考答案】B

【題號】question 172

【題目】

A solutions architect is creating a new Amazon CloudFront distribution for an application. Some of the information submitted by users is sensitive. The application uses HTTPS but needs another layer of security. The sensitive information should be protected throughout the entire application stack, and access to the information should be restricted to certain applications.

Which action should the solutions architect take?

解決方案架構師正在為應用程式建立新的 Amazon CloudFront 發行版。用戶提交的一些資訊屬於敏感資訊。該應用程式使用 HTTPS，但需要另一層安全性。敏感資訊應在整個應用程式堆疊中受到保護，並且對資訊的存取應僅限於某些應用程式。

解決方案架構師應該採取哪些行動？

【選項】

A. Configure a CloudFront signed URL.

B. Configure a CloudFront signed cookie.

C. Configure a CloudFront field-level encryption profile.

D.Configure CloudFront and set the Origin Protocol Policy setting to HTTPS Only for the Viewer Protocol Policy.

【參考答案】C

【題號】question 173

【題目】

A gaming company hosts a browser-based application on AWS. The users of the application consume a large number of videos and images that are stored in Amazon S3. This content is the same for all users.

The application has increased in popularity, and millions of users worldwide accessing these media files. The company wants to provide the files to the users while reducing the load on the origin.

Which solution meets these requirements MOST cost-effectively?

一家遊戲公司在 AWS 上託管基於瀏覽器的應用程式。該應用程式的用戶使用儲存在 Amazon S3 中的大量影片和圖像。該內容對於所有使用者都是相同的。

該應用程式越來越受歡迎，全球有數百萬用戶存取這些媒體檔案。該公司希望向用戶提供文件，同時減少來源的負載。

哪種解決方案最經濟有效率地滿足這些要求？

【選項】

- A.Deploy an AWS Global Accelerator accelerator in front of the web servers.
- B.Deploy an Amazon CloudFront web distribution in front of the S3 bucket.
- C.Deploy an Amazon ElastiCache for Redis instance in front of the web servers.
- D.Deploy an Amazon ElastiCache for Memcached instance in front of the web servers.

【參考答案】B

【題號】question 174

【題目】

A company has a multi-tier application that runs six front-end web servers in an Amazon EC2 Auto Scaling group in a single Availability Zone behind an Application Load Balancer (ALB). A solutions architect needs to modify the infrastructure to be highly available without modifying the application.

Which architecture should the solutions architect choose that provides high availability?

一家公司有一個多層應用程式，該應用程式在應用程式負載平衡器 (ALB) 後面的單一可用區中的 Amazon EC2 Auto Scaling 群組中運行六個前端 Web 伺服器。解決方案架構師需要修改基礎架構以使其具有高可用性，而無需修改應用程式。

解決方案架構師應該選擇哪種架構來提供高可用性？

【選項】

- A. Create an Auto Scaling group that uses three instances across each of two Regions.
- B. Modify the Auto Scaling group to use three instances across each of two Availability Zones.
- C. Create an Auto Scaling template that can be used to quickly create more instances in another Region.
- D. Change the ALB in front of the Amazon EC2 instances in a round-robin configuration to balance traffic to the web tier.

【參考答案】B

【題號】question 175

【題目】

An ecommerce company has an order-processing application that uses Amazon API Gateway and an AWS Lambda function. The application stores data in an Amazon Aurora PostgreSQL database. During a recent sales event, a sudden surge in customer orders

occurred. Some customers experienced timeouts, and the application did not process the orders of those customers.

A solutions architect determined that the CPU utilization and memory utilization were high on the database because of a large number of open connections. The solutions architect needs to prevent the timeout errors while making the least possible changes to the application.

Which solution will meet these requirements?

一家電子商務公司有一個使用 Amazon API Gateway 和 AWS Lambda 函數的訂單處理應用程式。該應用程式將資料儲存在 Amazon Aurora PostgreSQL 資料庫中。在最近的一次銷售活動中，客戶訂單突然激增。一些客戶遇到超時，應用程式無法處理這些客戶的訂單。

解決方案架構師確定，由於存在大量開啟的連接，資料庫的 CPU 使用率和記憶體使用率很高。解決方案架構師需要防止逾時錯誤，同時對應用程式進行盡可能少的更改。

哪種解決方案可以滿足這些要求？

【選項】

A. Configure provisioned concurrency for the Lambda function. Modify the database to be a global database in multiple AWS Regions.

B. Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint.

C. Create a read replica for the database in a different AWS Region. Use query string parameters in API Gateway to route traffic to the read replica.

D. Migrate the data from Aurora PostgreSQL to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). Modify the Lambda function to use the DynamoDB table.

【參考答案】B

【題號】question 176

【題目】

An application runs on Amazon EC2 instances in private subnets. The application needs to access an Amazon DynamoDB table.

What is the MOST secure way to access the table while ensuring that the traffic does not leave the AWS network?

應用程式在私有子網路中的 Amazon EC2 執行個體上執行。應用程式需要存取 Amazon DynamoDB 表。

在確保流量不會離開 AWS 網路的同時存取表的最安全方法是什麼？

【選項】

- A. Use a VPC endpoint for DynamoDB.
- B. Use a NAT gateway in a public subnet.
- C. Use a NAT instance in a private subnet.
- D. Use the internet gateway attached to the VPC.

【參考答案】A

【題號】question 177

【題目】

An entertainment company is using Amazon DynamoDB to store media metadata. The application is read intensive and experiencing delays. The company does not have staff to handle additional operational overhead and needs to improve the performance efficiency of DynamoDB without reconfiguring the application.

What should a solutions architect recommend to meet this requirement?

一家娛樂公司正在使用 Amazon DynamoDB 來儲存媒體元資料。該應用程式是讀取密集型的並且遇到延遲。該公司沒有員工來處理額外的營運開銷，需要在不重新配置應用程式的情況下提高 DynamoDB 的效能效率。

解決方案架構師應該建議什麼來滿足此要求？

【選項】

- A. Use Amazon ElastiCache for Redis.
- B. Use Amazon DynamoDB Accelerator (DAX).
- C. Replicate data by using DynamoDB global tables.
- D. Use Amazon ElastiCache for Memcached with Auto Discovery enabled.

【參考答案】B

【題號】question 178

【題目】

A company's infrastructure consists of Amazon EC2 instances and an Amazon RDS DB instance in a single AWS Region. The company wants to back up its data in a separate Region.

Which solution will meet these requirements with the LEAST operational overhead?

本公司的基礎架構由單一 AWS 區域中的 Amazon EC2 執行個體和 Amazon RDS 資料庫執行個體組成。該公司希望將其資料備份到一個單獨的區域。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A. Use AWS Backup to copy EC2 backups and RDS backups to the separate Region.
- B. Use Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region.
- C. Create Amazon Machine Images (AMIs) of the EC2 instances. Copy the AMIs to the separate Region. Create a read replica for the RDS DB instance in the separate Region.
- D. Create Amazon Elastic Block Store (Amazon EBS) snapshots. Copy the EBS snapshots to the separate Region. Create RDS snapshots. Export the RDS snapshots to Amazon S3. Configure S3 Cross-Region Replication (CRR) to the separate Region.

【參考答案】A

【題號】question 179

【題目】

A solutions architect needs to securely store a database user name and password that an application uses to access an Amazon RDS DB instance. The application that accesses the database runs on an Amazon EC2 instance. The solutions architect wants to create a secure parameter in AWS Systems Manager Parameter Store.

What should the solutions architect do to meet this requirement?

解決方案架構師需要安全地儲存應用程式用於存取 Amazon RDS 資料庫執行個體的資料庫使用者名稱和密碼。存取資料庫的應用程式在 Amazon EC2 執行個體上執行。解決方案架構師希望在 AWS Systems Manager Parameter Store 中建立安全參數。

解決方案架構師應該做什麼來滿足這個要求？

【選項】

A.Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance.

B.Create an IAM policy that allows read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM policy to the EC2 instance.

C.Create an IAM trust relationship between the Parameter Store parameter and the EC2 instance. Specify Amazon RDS as a principal in the trust policy.

D.Create an IAM trust relationship between the DB instance and the EC2 instance. Specify Systems Manager as a principal in the trust policy.

【參考答案】A

【題號】question 180

【題目】

A company is designing a cloud communications platform that is driven by APIs. The application is hosted on Amazon EC2 instances behind a Network Load Balancer (NLB). The company uses Amazon API Gateway to provide external users with access to the application through APIs. The company wants to protect the platform against web exploits like SQL injection and also wants to detect and mitigate large, sophisticated DDoS attacks.

Which combination of solutions provides the MOST protection? (Choose two.)

一家公司正在設計一個由 API 驅動的雲端通訊平台。該應用程式託管在網路負載平衡器 (NLB) 後面的 Amazon EC2 執行個體上。該公司使用 Amazon API Gateway 為外部用戶提供透過 API 存取應用程式的權限。該公司希望保護平台免受 SQL 注入等 Web 攻擊，還希望檢測和緩解大型、複雜的 DDoS 攻擊。

哪種解決方案組合可提供最強的保護？（選兩個。）

【選項】

- A. Use AWS WAF to protect the NLB.
- B. Use AWS Shield Advanced with the NLB.
- C. Use AWS WAF to protect Amazon API Gateway.
- D. Use Amazon GuardDuty with AWS Shield Standard
- E. Use AWS Shield Standard with Amazon API Gateway.

【參考答案】BC

【題號】question 181

【題目】

A company has a legacy data processing application that runs on Amazon EC2 instances. Data is processed sequentially, but the order of results does not matter. The application

uses a monolithic architecture. The only way that the company can scale the application to meet increased demand is to increase the size of the instances.

The company's developers have decided to rewrite the application to use a microservices architecture on Amazon Elastic Container Service (Amazon ECS).

What should a solutions architect recommend for communication between the microservices?

一家公司有一個在 Amazon EC2 執行個體上執行的舊資料處理應用程式。數據按順序處理，但結果的順序並不重要。該應用程式使用整體架構。公司擴展應用程式以滿足不斷增長的需求的唯一方法是增加實例的大小。

該公司的開發人員決定重寫該應用程式，以使用 Amazon Elastic Container Service (Amazon ECS) 上的微服務架構。

解決方案架構師應該為微服務之間的通訊推薦什麼？

【選項】

A.Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue.

B.Create an Amazon Simple Notification Service (Amazon SNS) topic. Add code to the data producers, and publish notifications to the topic. Add code to the data consumers to subscribe to the topic.

C.Create an AWS Lambda function to pass messages. Add code to the data producers to call the Lambda function with a data object. Add code to the data consumers to receive a data object that is passed from the Lambda function.

D.Create an Amazon DynamoDB table. Enable DynamoDB Streams. Add code to the data producers to insert data into the table. Add code to the data consumers to use the DynamoDB Streams API to detect new table entries and retrieve the data.

【參考答案】A

【題號】question 182

【題目】

A company wants to migrate its MySQL database from on premises to AWS. The company recently experienced a database outage that significantly impacted the business. To ensure this does not happen again, the company wants a reliable database solution on AWS that minimizes data loss and stores every transaction on at least two nodes.

Which solution meets these requirements?

一家公司希望將其 MySQL 資料庫從本地端遷移到 AWS。該公司最近經歷了一次資料庫中斷，對業務產生了重大影響。為了確保這種情況不再發生，該公司希望在 AWS 上有一個可靠的資料庫解決方案，最大限度地減少資料遺失並將每筆交易儲存在至少兩個節點上。

哪種解決方案滿足這些要求？

【選項】

A.Create an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones.

B.Create an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data.

C.Create an Amazon RDS MySQL DB instance and then create a read replica in a separate AWS Region that synchronously replicates the data.

D.Create an Amazon EC2 instance with a MySQL engine installed that triggers an AWS Lambda function to synchronously replicate the data to an Amazon RDS MySQL DB instance.

【參考答案】B

【題號】question 183

【題目】

A company is building a new dynamic ordering website. The company wants to minimize server maintenance and patching. The website must be highly available and must scale read and write capacity as quickly as possible to meet changes in user demand.

Which solution will meet these requirements?

一家公司正在建立一個新的動態訂購網站。該公司希望最大限度地減少伺服器維護和修補。網站必須具有高可用性，並且必須盡快擴展讀寫容量，以滿足使用者需求的變化。

哪種解決方案可以滿足這些要求？

【選項】

A.Host static content in Amazon S3. Host dynamic content by using Amazon API Gateway and AWS Lambda. Use Amazon DynamoDB with on-demand capacity for the database. Configure Amazon CloudFront to deliver the website content.

B.Host static content in Amazon S3. Host dynamic content by using Amazon API Gateway and AWS Lambda. Use Amazon Aurora with Aurora Auto Scaling for the database. Configure Amazon CloudFront to deliver the website content.

C.Host all the website content on Amazon EC2 instances. Create an Auto Scaling group to scale the EC2 instances. Use an Application Load Balancer to distribute traffic. Use Amazon DynamoDB with provisioned write capacity for the database.

D.Host all the website content on Amazon EC2 instances. Create an Auto Scaling group to scale the EC2 instances. Use an Application Load Balancer to distribute traffic. Use Amazon Aurora with Aurora Auto Scaling for the database.

【參考答案】A

【題號】question 184

【題目】

A company has an AWS account used for software engineering. The AWS account has access to the company's on-premises data center through a pair of AWS Direct Connect connections. All non-VPC traffic routes to the virtual private gateway.

A development team recently created an AWS Lambda function through the console. The development team needs to allow the function to access a database that runs in a private subnet in the company's data center.

Which solution will meet these requirements?

一家公司擁有一個用於軟體工程的 AWS 帳戶。AWS 帳戶可以透過一對 AWS Direct Connect 連線存取公司的本機資料中心。所有非 VPC 流量都會路由到虛擬專用網關。

一個開發團隊最近透過控制台創建了一個 AWS Lambda 函數。開發團隊需要允許該函數存取在公司資料中心的私有子網路中運行的資料庫。

哪種解決方案可以滿足這些要求？

【選項】

- A. Configure the Lambda function to run in the VPC with the appropriate security group.
- B. Set up a VPN connection from AWS to the data center. Route the traffic from the Lambda function through the VPN.
- C. Update the route tables in the VPC to allow the Lambda function to access the on-premises data center through Direct Connect.
- D. Create an Elastic IP address. Configure the Lambda function to send traffic through the Elastic IP address without an elastic network interface.

【參考答案】A

【題號】question 185

【題目】

A company runs an application using Amazon ECS. The application creates resized versions of an original image and then makes Amazon S3 API calls to store the resized images in Amazon S3.

How can a solutions architect ensure that the application has permission to access Amazon S3?

一家公司使用 Amazon ECS 運行應用程式。該應用程式創建原始圖像的調整大小版本，然後進行 Amazon S3 API 呼叫以將調整大小的圖像儲存在 Amazon S3 中。

解決方案架構師如何確保應用程式有權存取 Amazon S3？

【選項】

- A. Update the S3 role in AWS IAM to allow read/write access from Amazon ECS, and then relaunch the container.
- B. Create an IAM role with S3 permissions, and then specify that role as the taskRoleArn in the task definition.
- C. Create a security group that allows access from Amazon ECS to Amazon S3, and update the launch configuration used by the ECS cluster.
- D. Create an IAM user with S3 permissions, and then relaunch the Amazon EC2 instances for the ECS cluster while logged in as this account.

【參考答案】B

【題號】question 186

【題目】

A company has a Windows-based application that must be migrated to AWS. The application requires the use of a shared Windows file system attached to multiple Amazon EC2 Windows instances that are deployed across multiple Availability Zone:

What should a solutions architect do to meet this requirement?

一家公司有一個基於 Windows 的應用程式，必須遷移到 AWS。該應用程式需要使用附加到跨多個可用區部署的多個 Amazon EC2 Windows 執行個體的共用 Windows 檔案系統：

解決方案架構師應該怎麼做才能滿足這項要求？

【選項】

- A. Configure AWS Storage Gateway in volume gateway mode. Mount the volume to each Windows instance.

B.Configure Amazon FSx for Windows File Server. Mount the Amazon FSx file system to each Windows instance.

C.Configure a file system by using Amazon Elastic File System (Amazon EFS). Mount the EFS file system to each Windows instance.

D.Configure an Amazon Elastic Block Store (Amazon EBS) volume with the required size. Attach each EC2 instance to the volume. Mount the file system within the volume to each Windows instance.

【參考答案】B

【題號】question 187

【題目】

A company is developing an ecommerce application that will consist of a load-balanced front end, a container-based application, and a relational database. A solutions architect needs to create a highly available solution that operates with as little manual intervention as possible.

Which solutions meet these requirements? (Choose two.)

一家公司正在開發一個電子商務應用程序，該應用程式由負載平衡前端、基於容器的應用程式和關聯式資料庫組成。解決方案架構師需要創建一個高度可用的解決方案，該解決方案在運行時需要盡可能少的手動幹預。

哪些解決方案滿足這些要求？（選兩個。）

【選項】

A.Create an Amazon RDS DB instance in Multi-AZ mode.

B.Create an Amazon RDS DB instance and one or more replicas in another Availability Zone.

C.Create an Amazon EC2 instance-based Docker cluster to handle the dynamic application load.

D.Create an Amazon Elastic Container Service (Amazon ECS) cluster with a Fargate launch type to handle the dynamic application load.

E.Create an Amazon Elastic Container Service (Amazon ECS) cluster with an Amazon EC2 launch type to handle the dynamic application load.

【參考答案】AD

【題號】question 188

【題目】

A company uses Amazon S3 as its data lake. The company has a new partner that must use SFTP to upload data files. A solutions architect needs to implement a highly available SFTP solution that minimizes operational overhead.

Which solution will meet these requirements?

一家公司使用 Amazon S3 作為其資料湖。該公司有一個新的合作夥伴，必須使用 SFTP 上傳資料檔案。解決方案架構師需要實施高度可用的 SFTP 解決方案，以最大限度地減少營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A.Use AWS Transfer Family to configure an SFTP-enabled server with a publicly accessible endpoint. Choose the S3 data lake as the destination.

B.Use Amazon S3 File Gateway as an SFTP server. Expose the S3 File Gateway endpoint URL to the new partner. Share the S3 File Gateway endpoint with the new partner.

C.Launch an Amazon EC2 instance in a private subnet in a VPC. Instruct the new partner to upload files to the EC2 instance by using a VPN. Run a cron job script, on the EC2 instance to upload files to the S3 data lake.

D.Launch Amazon EC2 instances in a private subnet in a VPC. Place a Network Load Balancer (NLB) in front of the EC2 instances. Create an SFTP listener port for the NLB. Share the NLB hostname with the new partner. Run a cron job script on the EC2 instances to upload files to the S3 data lake.

【參考答案】A

【題號】question 189

【題目】

A company needs to store contract documents. A contract lasts for 5 years. During the 5-year period, the company must ensure that the documents cannot be overwritten or deleted. The company needs to encrypt the documents at rest and rotate the encryption keys automatically every year.

Which combination of steps should a solutions architect take to meet these requirements with the LEAST operational overhead? (Choose two.)

公司需要儲存合約文件。合約期限為 5 年。5年期間，公司必須確保文件不會被覆蓋或刪除。該公司需要對靜態文件進行加密，並每年自動輪換加密金鑰。

解決方案架構師應採取哪種步驟組合才能以最少的營運開銷滿足這些要求？（選兩個。）

【選項】

- A.Store the documents in Amazon S3. Use S3 Object Lock in governance mode.
- B.Store the documents in Amazon S3. Use S3 Object Lock in compliance mode.
- C.Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Configure key rotation.
- D.Use server-side encryption with AWS Key Management Service (AWS KMS) customer managed keys. Configure key rotation.
- E.Use server-side encryption with AWS Key Management Service (AWS KMS) customer provided (imported) keys. Configure key rotation.

【參考答案】BD

【題號】question 190

【題目】

A company has a web application that is based on Java and PHP. The company plans to move the application from on premises to AWS. The company needs the ability to test new site features frequently. The company also needs a highly available and managed solution that requires minimum operational overhead.

Which solution will meet these requirements?

一家公司有一個基於 Java 和 PHP 的 Web 應用程式。該公司計劃將應用程式從本地遷移到 AWS。該公司需要能夠經常測試新網站功能。該公司還需要一個高度可用且託管的解決方案，並且需要最小的營運開銷。

哪種解決方案可以滿足這些要求？

【選項】

A.Create an Amazon S3 bucket. Enable static web hosting on the S3 bucket. Upload the static content to the S3 bucket. Use AWS Lambda to process all dynamic content.

B.Deploy the web application to an AWS Elastic Beanstalk environment. Use URL swapping to switch between multiple Elastic Beanstalk environments for feature testing.

C.Deploy the web application to Amazon EC2 instances that are configured with Java and PHP. Use Auto Scaling groups and an Application Load Balancer to manage the website's availability.

D.Containerize the web application. Deploy the web application to Amazon EC2 instances. Use the AWS Load Balancer Controller to dynamically route traffic between containers that contain the new site features for testing.

【參考答案】B

【題號】question 191

【題目】

A company has an ordering application that stores customer information in Amazon RDS for MySQL. During regular business hours, employees run one-time queries for reporting purposes. Timeouts are occurring during order processing because the reporting queries are taking a long time to run. The company needs to eliminate the timeouts without preventing employees from performing queries.

What should a solutions architect do to meet these requirements?

一家公司有一個訂購應用程式，該應用程式將客戶資訊儲存在 Amazon RDS for MySQL 中。在正常工作時間內，員工會出於報告目的執行一次性查詢。訂單處理期間發生逾時，因為報表查詢需要很長時間才能運行。公司需要在不阻止員工執行查詢的情況下消除逾時。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A.Create a read replica. Move reporting queries to the read replica.
- B.Create a read replica. Distribute the ordering application to the primary DB instance and the read replica.
- C.Migrate the ordering application to Amazon DynamoDB with on-demand capacity.
- D.Schedule the reporting queries for non-peak hours.

【參考答案】A

【題號】question 192

【題目】

A hospital wants to create digital copies for its large collection of historical written records. The hospital will continue to add hundreds of new documents each day. The hospital's data team will scan the documents and will upload the documents to the AWS Cloud.

A solutions architect must implement a solution to analyze the documents, extract the medical information, and store the documents so that an application can run SQL queries on the data. The solution must maximize scalability and operational efficiency.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

一家醫院希望為其大量歷史書面記錄建立數位副本。醫院每天將繼續增加數百份新文件。醫院的數據團隊將掃描文件並將文件上傳到AWS雲端。

解決方案架構師必須實作一個解決方案來分析文件、提取醫療資訊並儲存文檔，以便應用程式可以對資料執行 SQL 查詢。此解決方案必須最大限度地提高可擴展性和營運效率。

解決方案架構師應該採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

A. Write the document information to an Amazon EC2 instance that runs a MySQL database.

B. Write the document information to an Amazon S3 bucket. Use Amazon Athena to query the data.

C. Create an Auto Scaling group of Amazon EC2 instances to run a custom application that processes the scanned files and extracts the medical information.

D. Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Rekognition to convert the documents to raw text. Use Amazon Transcribe Medical to detect and extract relevant medical information from the text.

E. Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Textract to convert the documents to raw text. Use Amazon Comprehend Medical to detect and extract relevant medical information from the text.

【參考答案】BE

【題號】question 193

【題目】

A company is running a batch application on Amazon EC2 instances. The application consists of a backend with multiple Amazon RDS databases. The application is causing a high number of reads on the databases. A solutions architect must reduce the number of database reads while ensuring high availability.

What should the solutions architect do to meet this requirement?

一家公司正在 Amazon EC2 執行個體上執行批次應用程式。該應用程式由具有多個 Amazon RDS 資料庫的後端組成。該應用程式導致對資料庫的大量讀取。解決方案架構師必須減少資料庫讀取次數，同時確保高可用性。

解決方案架構師應該做什麼來滿足這個要求？

【選項】

- A.Add Amazon RDS read replicas.
- B.Use Amazon ElastiCache for Redis.
- C.Use Amazon Route 53 DNS caching
- D.Use Amazon ElastiCache for Memcached.

【參考答案】A

【題號】question 194

【題目】

A company needs to run a critical application on AWS. The company needs to use Amazon EC2 for the application's database. The database must be highly available and must fail over automatically if a disruptive event occurs.

Which solution will meet these requirements?

一家公司需要在 AWS 上運行關鍵應用程式。該公司需要使用 Amazon EC2 作為應用程式的資料庫。資料庫必須具有高可用性，並且在發生破壞性事件時必須自動進行故障轉移。

哪種解決方案可以滿足這些要求？

【選項】

- A.Launch two EC2 instances, each in a different Availability Zone in the same AWS Region. Install the database on both EC2 instances. Configure the EC2 instances as a cluster. Set up database replication.
- B.Launch an EC2 instance in an Availability Zone. Install the database on the EC2 instance. Use an Amazon Machine Image (AMI) to back up the data. Use AWS CloudFormation to automate provisioning of the EC2 instance if a disruptive event occurs.

C.Launch two EC2 instances, each in a different AWS Region. Install the database on both EC2 instances. Set up database replication. Fail over the database to a second Region.

D.Launch an EC2 instance in an Availability Zone. Install the database on the EC2 instance. Use an Amazon Machine Image (AMI) to back up the data. Use EC2 automatic recovery to recover the instance if a disruptive event occurs.

【參考答案】A

【題號】question 195

【題目】

A company's order system sends requests from clients to Amazon EC2 instances. The EC2 instances process the orders and then store the orders in a database on Amazon RDS. Users report that they must reprocess orders when the system fails. The company wants a resilient solution that can process orders automatically if a system outage occurs.

What should a solutions architect do to meet these requirements?

本公司的訂單系統將客戶端的請求傳送到 Amazon EC2 執行個體。EC2 執行個體處理訂單，然後將訂單儲存在 Amazon RDS 上的資料庫中。用戶報告說，當系統出現故障時，他們必須重新處理訂單。該公司需要一種彈性解決方案，能夠在系統發生故障時自動處理訂單。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A.Move the EC2 instances into an Auto Scaling group. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to target an Amazon Elastic Container Service (Amazon ECS) task.

B.Move the EC2 instances into an Auto Scaling group behind an Application Load Balancer (ALB). Update the order system to send messages to the ALB endpoint.

C.Move the EC2 instances into an Auto Scaling group. Configure the order system to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the EC2 instances to consume messages from the queue.

D.Create an Amazon Simple Notification Service (Amazon SNS) topic. Create an AWS Lambda function, and subscribe the function to the SNS topic. Configure the order system to send messages to the SNS topic. Send a command to the EC2 instances to process the messages by using AWS Systems Manager Run Command.

【參考答案】C

【題號】question 196

【題目】

A company runs an application on a large fleet of Amazon EC2 instances. The application reads and writes entries into an Amazon DynamoDB table. The size of the DynamoDB table continuously grows, but the application needs only data from the last 30 days. The company needs a solution that minimizes cost and development effort.

Which solution meets these requirements?

一家公司在大量 Amazon EC2 執行個體上執行應用程式。該應用程式讀取 Amazon DynamoDB 表中的條目並將其寫入其中。DynamoDB 表的大小不斷增長，但應用程式只需要最近 30 天的資料。該公司需要一種能夠最大限度地減少成本和開發工作的解決方案。

哪種解決方案滿足這些要求？

【選項】

A.Use an AWS CloudFormation template to deploy the complete solution. Redeploy the CloudFormation stack every 30 days, and delete the original stack.

B.Use an EC2 instance that runs a monitoring application from AWS Marketplace. Configure the monitoring application to use Amazon DynamoDB Streams to store the timestamp when a new item is created in the table. Use a script that runs on the EC2 instance to delete items that have a timestamp that is older than 30 days.

C.Configure Amazon DynamoDB Streams to invoke an AWS Lambda function when a new item is created in the table. Configure the Lambda function to delete items in the table that are older than 30 days.

D.Extend the application to add an attribute that has a value of the current timestamp plus 30 days to each new item that is created in the table. Configure DynamoDB to use the attribute as the TTL attribute.

【參考答案】D

【題號】question 197

【題目】

A company has a Microsoft .NET application that runs on an on-premises Windows Server. The application stores data by using an Oracle Database Standard Edition server. The company is planning a migration to AWS and wants to minimize development changes while moving the application. The AWS application environment should be highly available.

Which combination of actions should the company take to meet these requirements? (Choose two.)

一家公司有一個在本機 Windows Server 上執行的 Microsoft .NET 應用程式。該應用程式使用 Oracle 資料庫標準版伺服器儲存資料。該公司正在計劃遷移到 AWS，並希望在遷移應用程式時最大程度地減少開發變更。AWS 應用程式環境應該是高可用的。

公司應該採取哪些行動組合來滿足這些要求？（選兩個。）

【選項】

A.Refactor the application as serverless with AWS Lambda functions running .NET Core.

B.Rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment.

C.Replatform the application to run on Amazon EC2 with the Amazon Linux Amazon Machine Image (AMI).

D.Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Amazon DynamoDB in a Multi-AZ deployment.

E.Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment.

【參考答案】BE

【題號】question 198

【題目】

A company runs a containerized application on a Kubernetes cluster in an on-premises data center. The company is using a MongoDB database for data storage. The company wants to migrate some of these environments to AWS, but no code changes or deployment method changes are possible at this time. The company needs a solution that minimizes operational overhead.

Which solution meets these requirements?

一家公司在本地資料中心的 Kubernetes 叢集上運行容器化應用程式。該公司正在使用 MongoDB 資料庫進行資料儲存。該公司希望將其中一些環境遷移到 AWS，但目前無法更改程式碼或部署方法。該公司需要一種能夠最大限度地減少營運開銷的解決方案。

哪種解決方案滿足這些要求？

【選項】

A. Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 worker nodes for compute and MongoDB on EC2 for data storage.

B. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate for compute and Amazon DynamoDB for data storage

C. Use Amazon Elastic Kubernetes Service (Amazon EKS) with Amazon EC2 worker nodes for compute and Amazon DynamoDB for data storage.

D. Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate for compute and Amazon DocumentDB (with MongoDB compatibility) for data storage.

【參考答案】D

【題號】question 199

【題目】

A telemarketing company is designing its customer call center functionality on AWS. The company needs a solution that provides multiple speaker recognition and generates transcript files. The company wants to query the transcript files to analyze the business patterns. The transcript files must be stored for 7 years for auditing purposes.

Which solution will meet these requirements?

一家電話行銷公司正在 AWS 上設計其客戶呼叫中心功能。該公司需要一種能夠提供多個說話者識別並產生轉錄檔案的解決方案。該公司想要查詢記錄文件以分析業務模式。成績單文件必須保存 7 年以供審計之用。

哪種解決方案可以滿足這些要求？

【選項】

A. Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use machine learning models for transcript file analysis.

B. Use Amazon Transcribe for multiple speaker recognition. Use Amazon Athena for transcript file analysis.

C. Use Amazon Translate for multiple speaker recognition. Store the transcript files in Amazon Redshift. Use SQL queries for transcript file analysis.

D. Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use Amazon Textract for transcript file analysis.

【參考答案】B

【題號】question 200

【題目】

A company hosts its application on AWS. The company uses Amazon Cognito to manage users. When users log in to the application, the application fetches required data from Amazon DynamoDB by using a REST API that is hosted in Amazon API Gateway. The

company wants an AWS managed solution that will control access to the REST API to reduce development efforts.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司在 AWS 上託管其應用程式。該公司使用 Amazon Cognito 來管理用戶。當使用者登入應用程式時，應用程式使用 Amazon API Gateway 中託管的 REST API 從 Amazon DynamoDB 取得所需資料。該公司需要一個 AWS 託管解決方案來控制對 REST API 的訪問，以減少開發工作。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A. Configure an AWS Lambda function to be an authorizer in API Gateway to validate which user made the request.
- B. For each user, create and assign an API key that must be sent with each request. Validate the key by using an AWS Lambda function.
- C. Send the user's email address in the header with every request. Invoke an AWS Lambda function to validate that the user with that email address has proper access.
- D. Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.

【參考答案】D
